

Artificial Intelligence for medical image analysis: Applications and challenges

Prof. Farida Cheriet Ait Hamou, Ing., Ph.D

Department of computer and software Engineering
Polytechnique Montréal

4th edition of the Women in Computer Science Workshop (WiSe)
December 15th, 2025

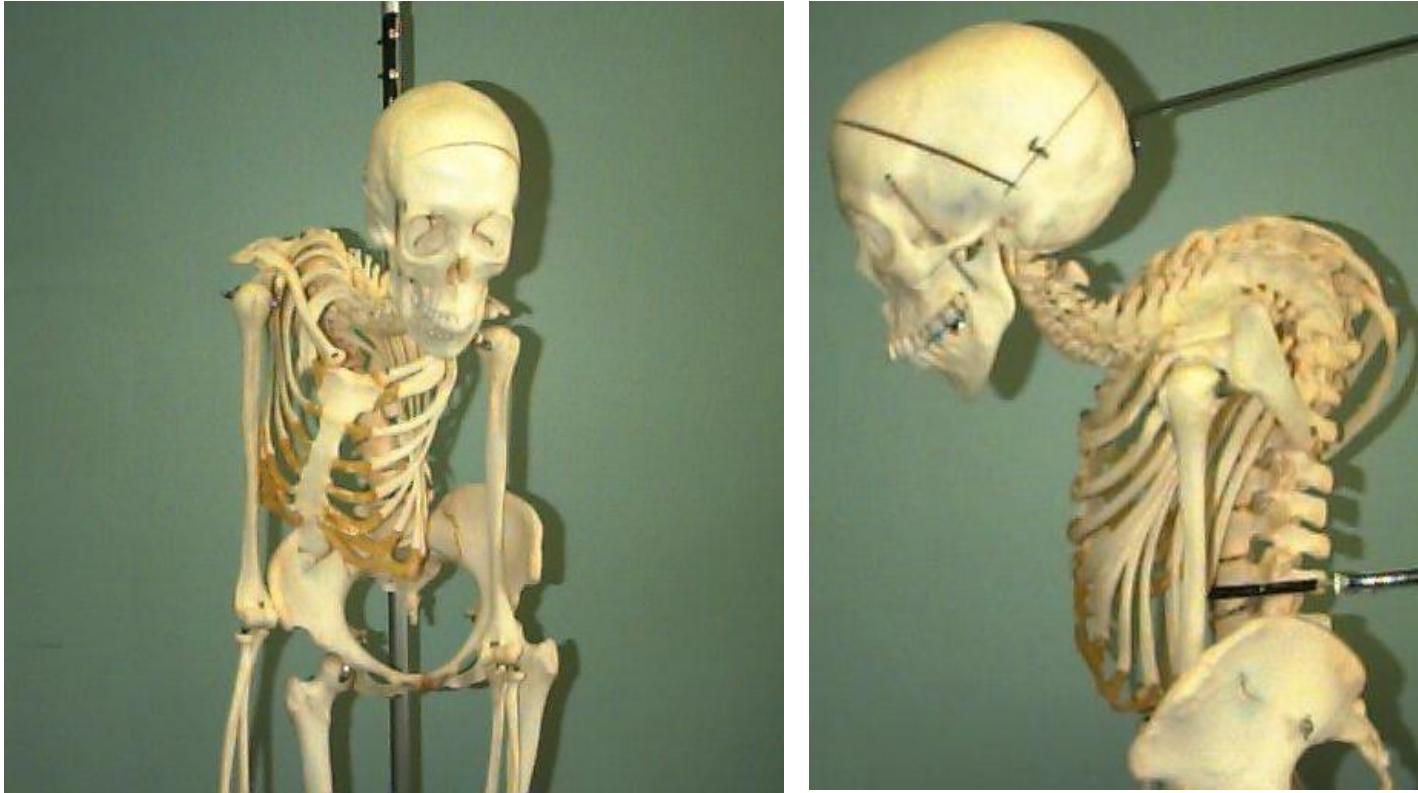


Research activities since 1999

- Orthopedics: Scoliosis deformities modeling
 - Cardiology: 3D reconstruction of vessels
 - Ophthalmology: Automatic diagnosis of ocular diseases
 - Others: Brain imaging, automatic ARDS diagnosis from X-ray images, prediction of ICU readmission and digital dentistry.
-
- **Most important achievement:**
 - 60 research master students
 - 50 PhD students
 - 15 postdoctoral fellows
 - 3 Research associates



AI in scoliosis deformities assessment



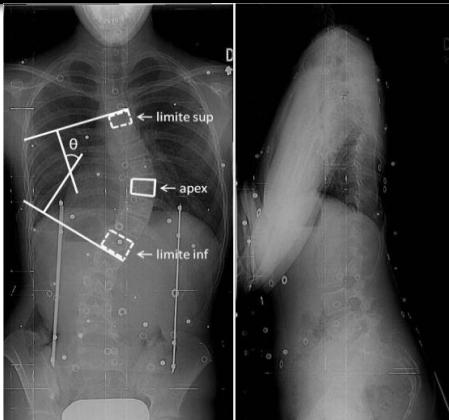
Specimen of Mary offered by Marc Asher to LIS3D



Scoliosis management

- Depending on the curve severity, in terms of Cobb angle

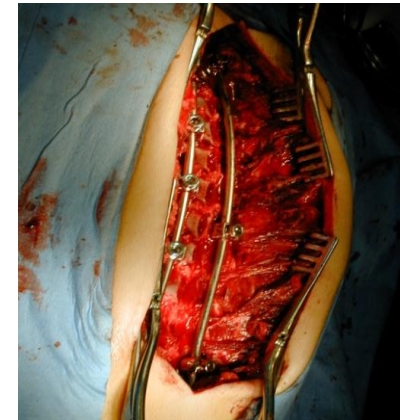
6 months follow-up



Brace



Spinal surgery



- 3D deformity of the spine and the rib cage
- Trunk deformities : rib hump, imbalance, asymmetries
 - **first signs of scoliosis**
 - **patients major concern** (*Asher, 2004*)

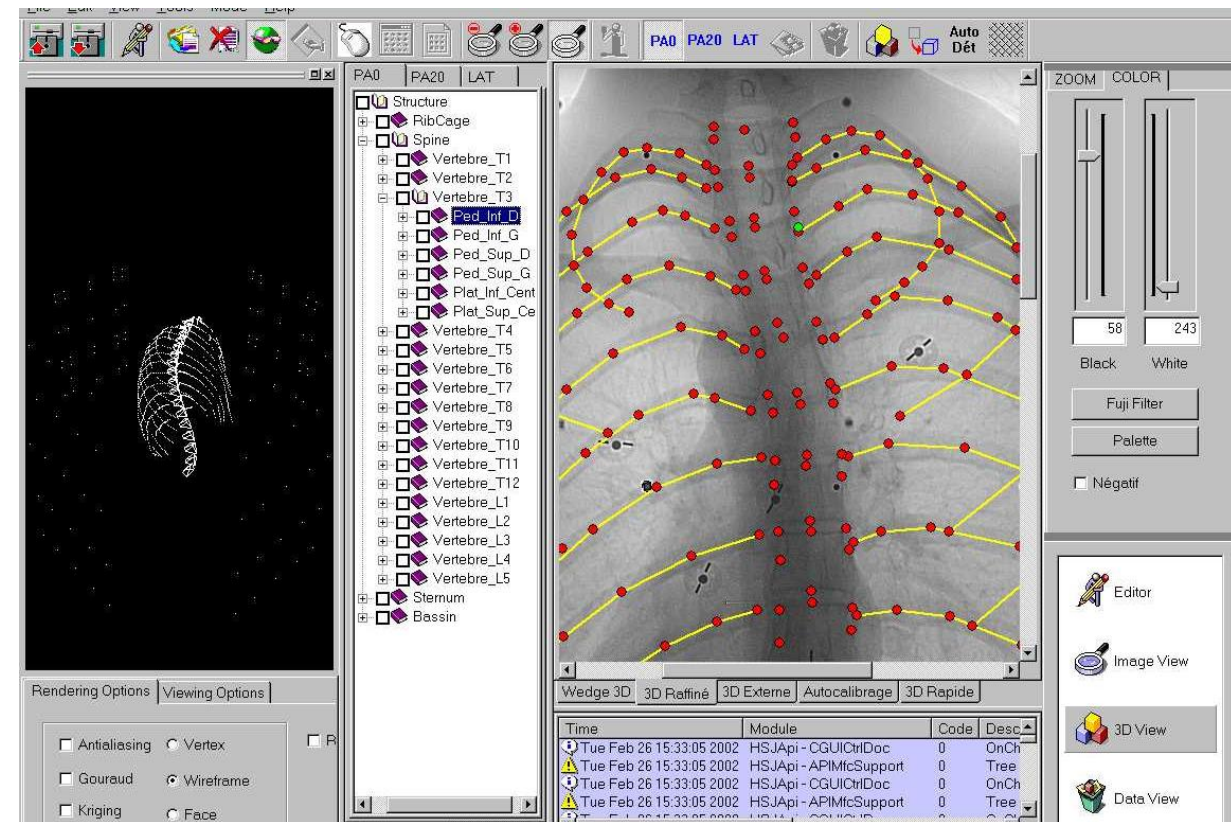
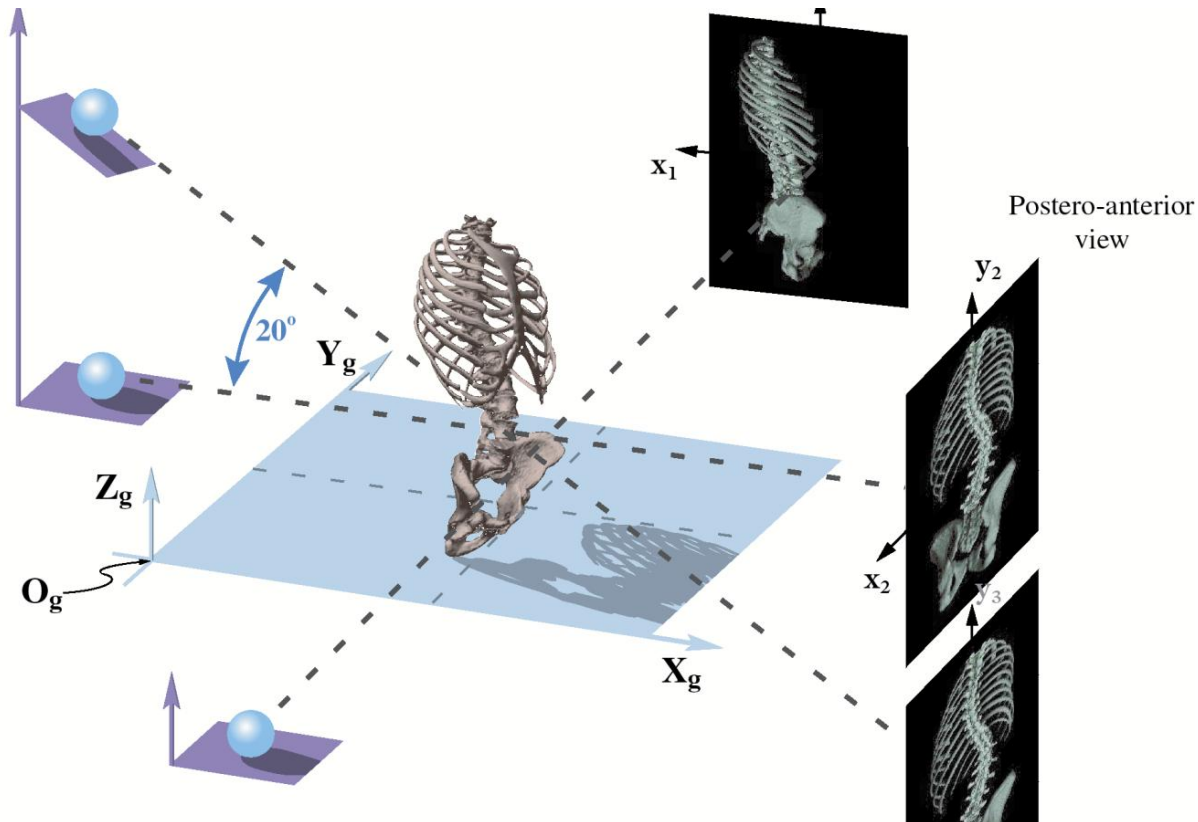


Explicit calibration of X-ray acquisition

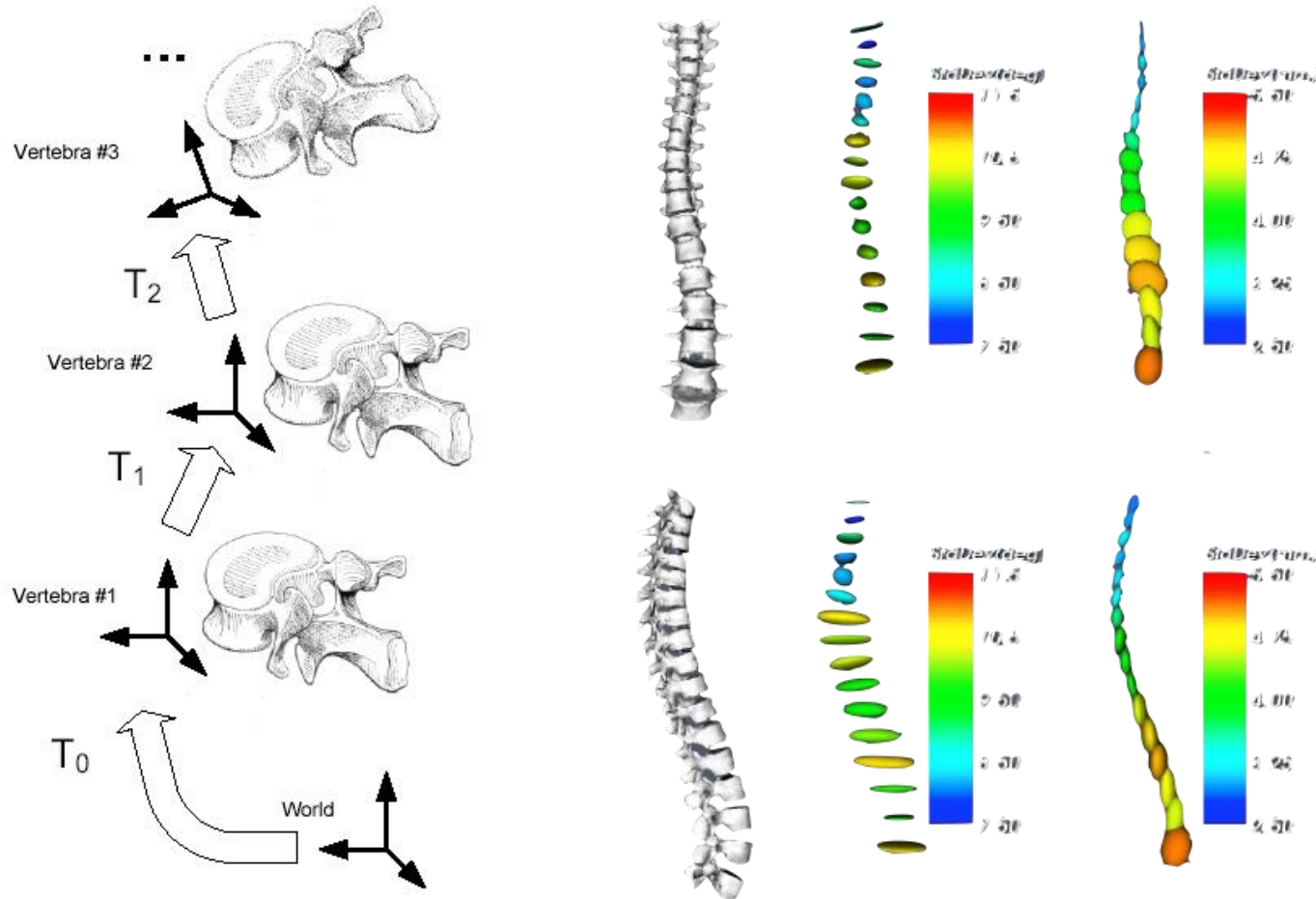
- Since 1993, a 3D reconstruction system of the rib cage is used
- 3D reconstruction is done from 2D X-rays
- More than 6000 patients have used this tool
- Problem of patient displacement between two acquisitions



3D scoliosis software



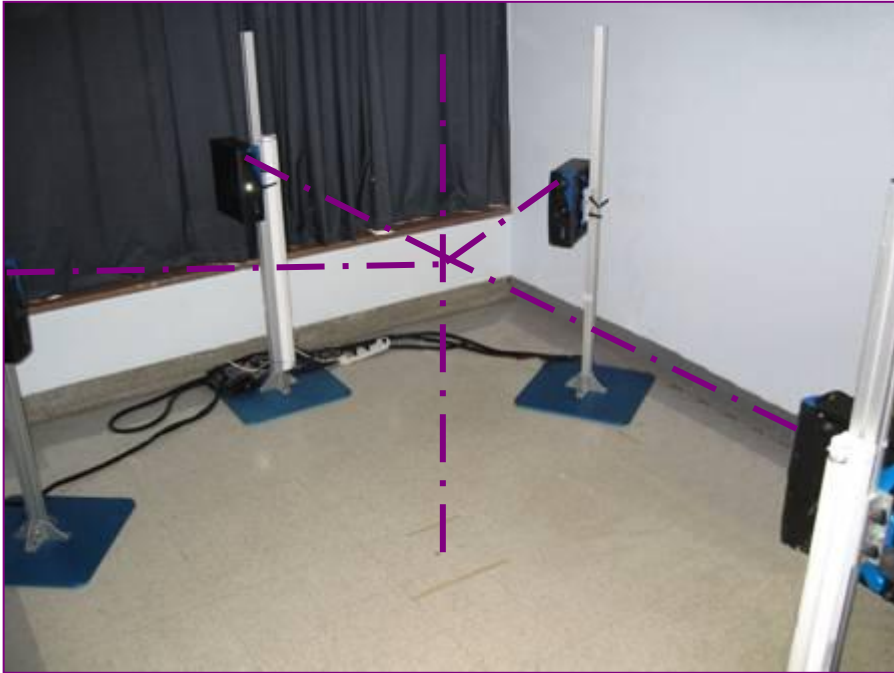
Articulated model of the spine



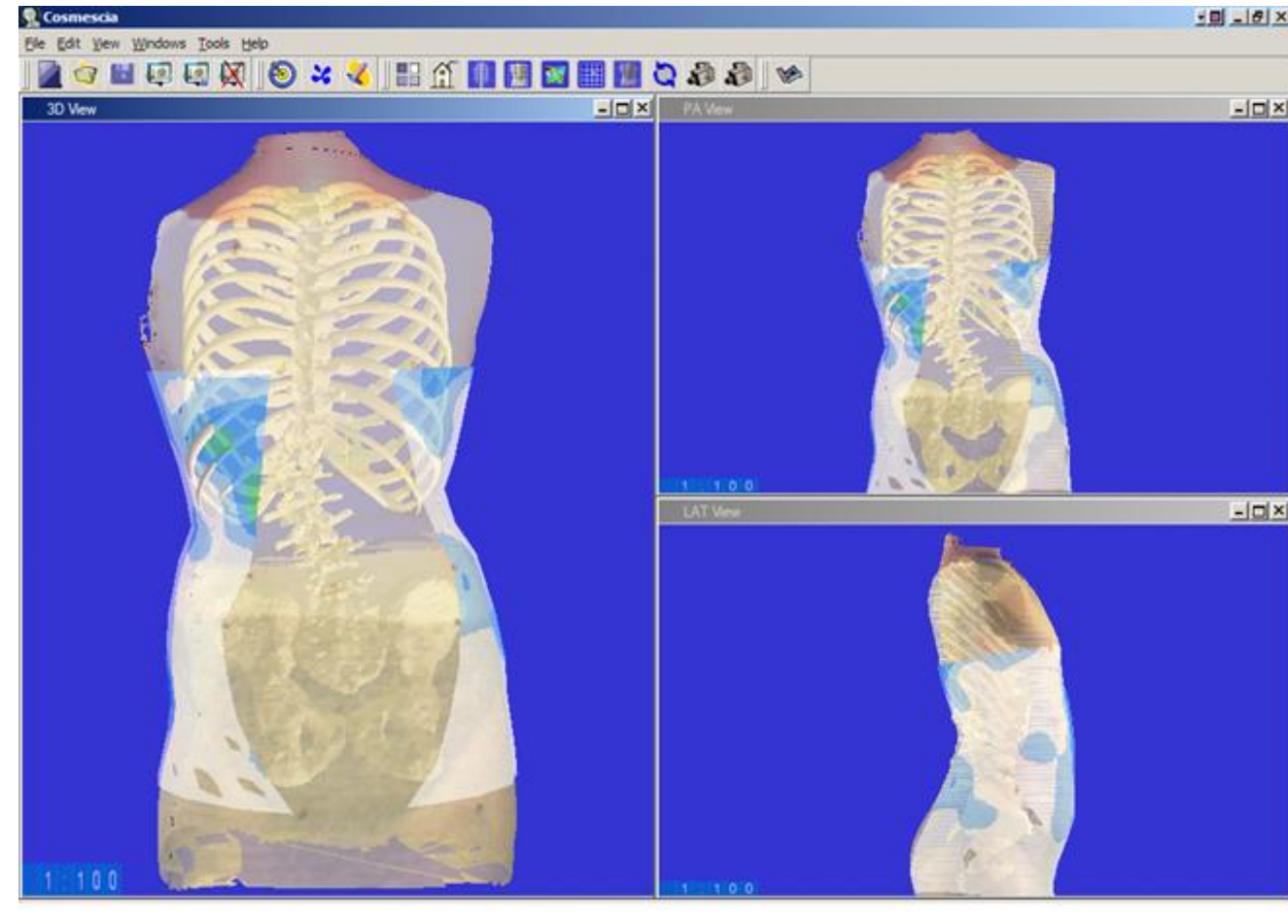
- Healthy mean shape.
 - Small right curvature in the frontal plane
- High variability of the translation along the vertical axis
 - Growth.
- High variability of the rotation along the postero-anterior axis

3D digitizing system (CFI grant)

- Non invasive follow-up of patients
- Relationship between external asymmetries of the trunk and bone deformities
- Brace design

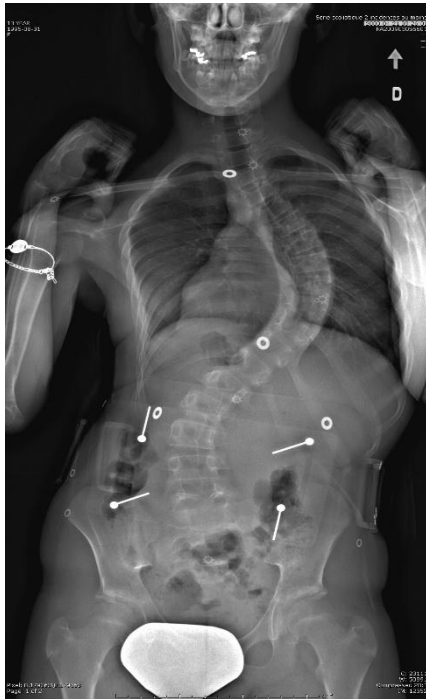


Lateral resolution 1.2 mm
Depth resolution 1.0 mm
Field of view 1185 mm x 889 mm

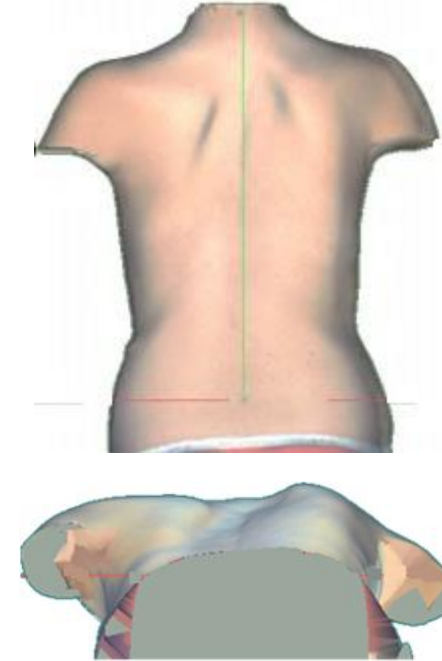


Surgical outcome

Before surgery



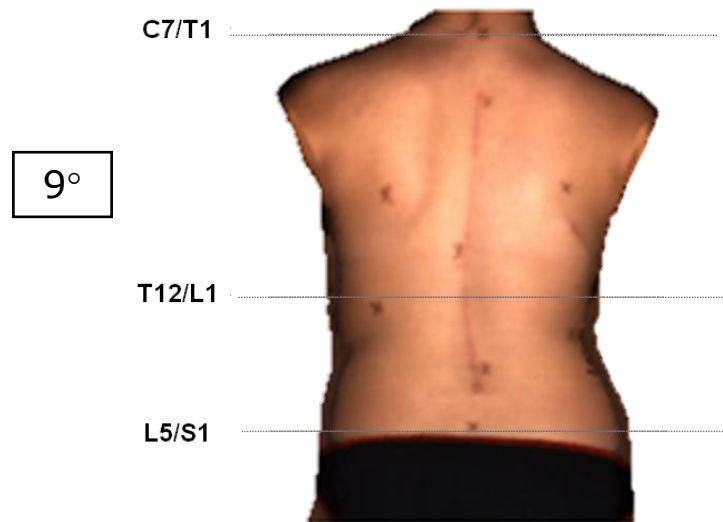
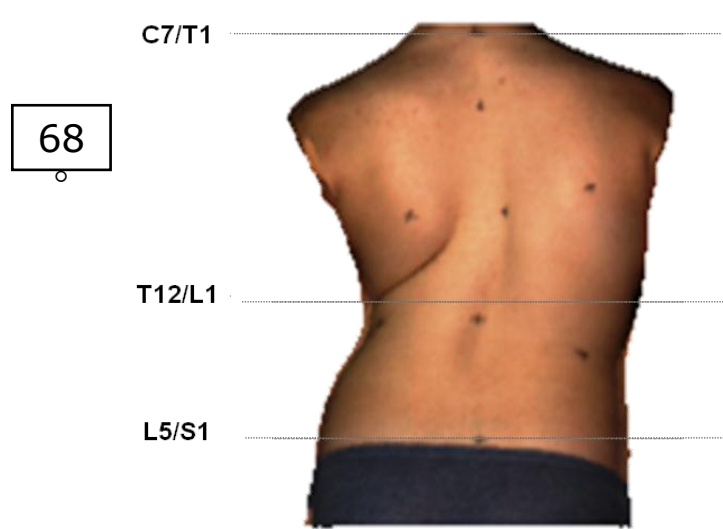
30 months after surgery



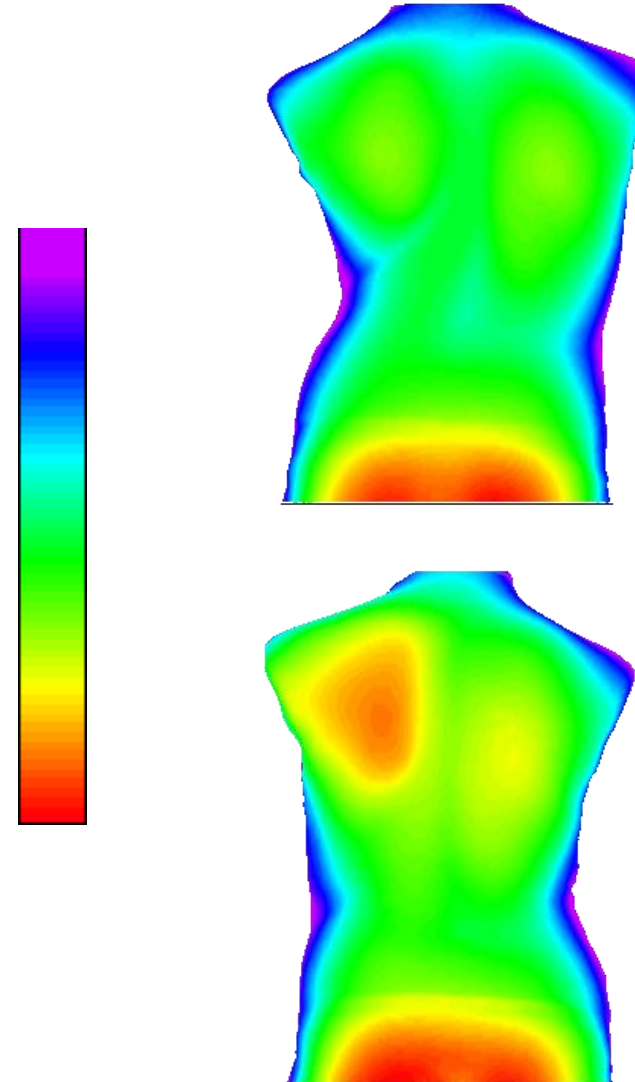
- Residual trunk deformities after treatment
- Patients dissatisfied after surgery (*Asher, 2004*)



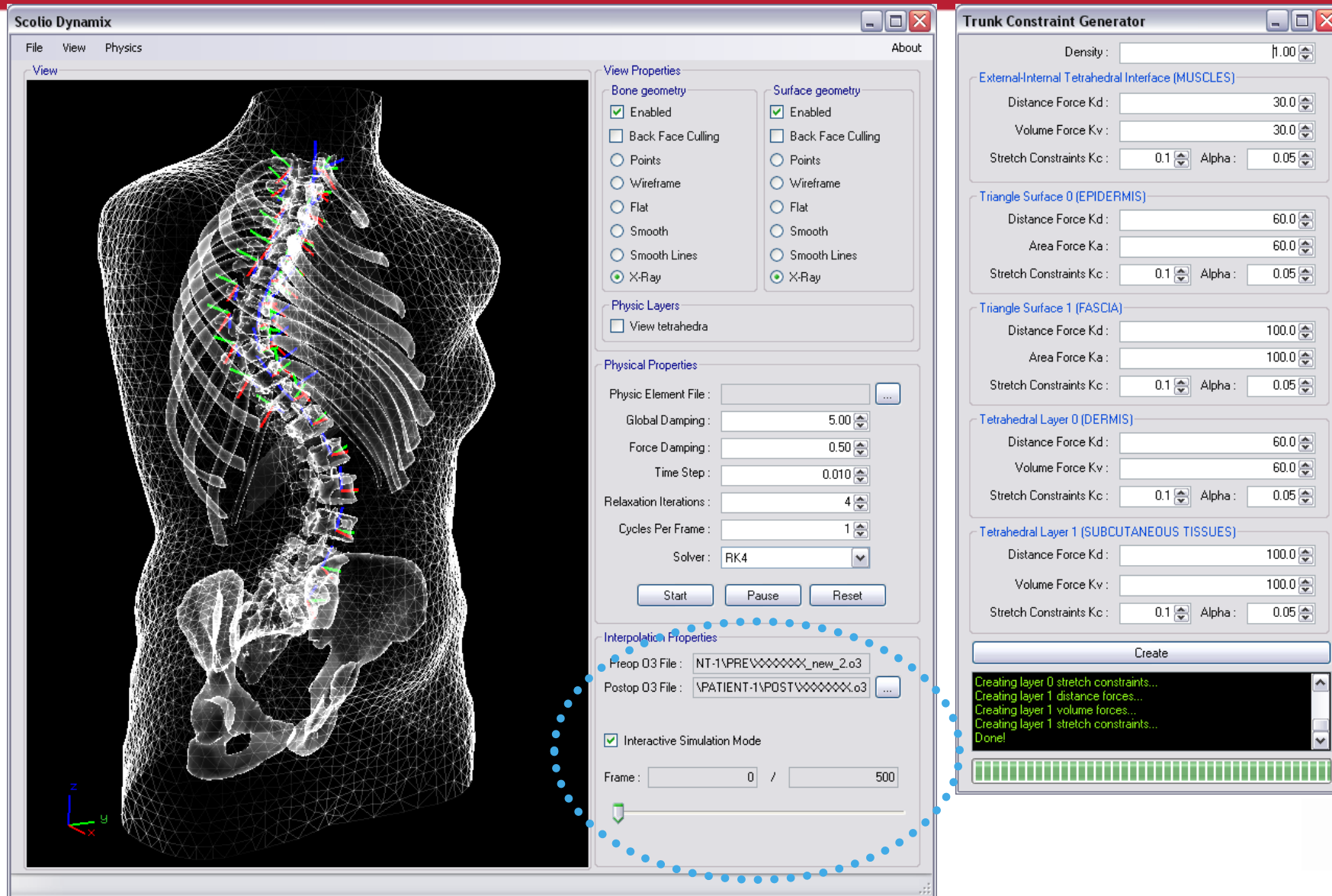
Worst correction of the rib hump



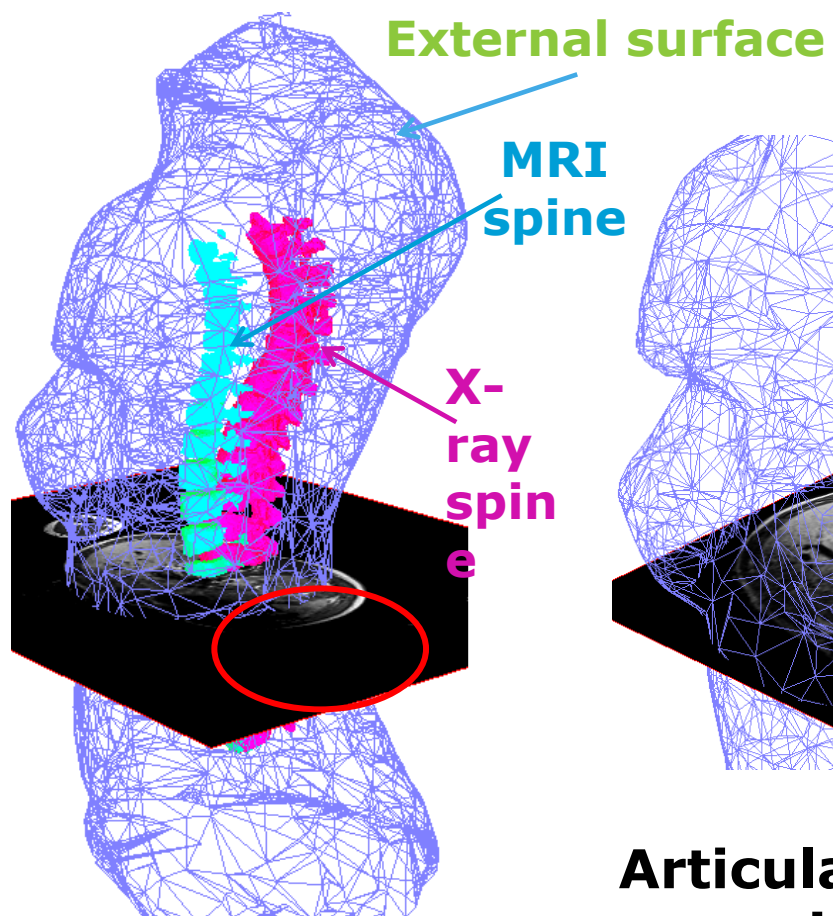
Postop



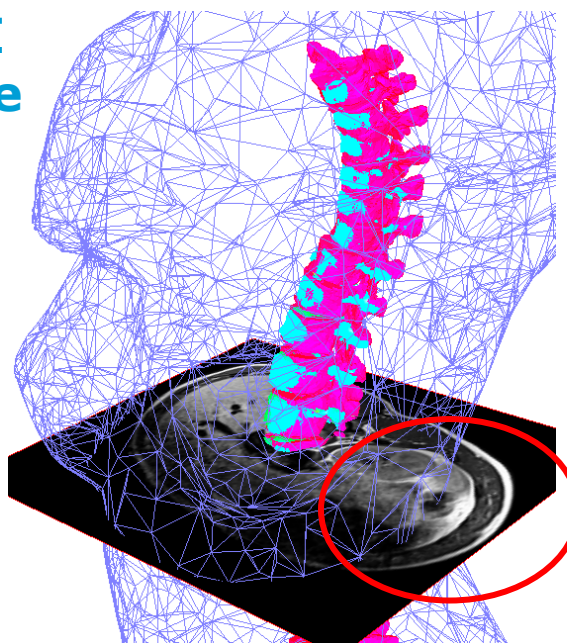
Postoperative trunk simulation



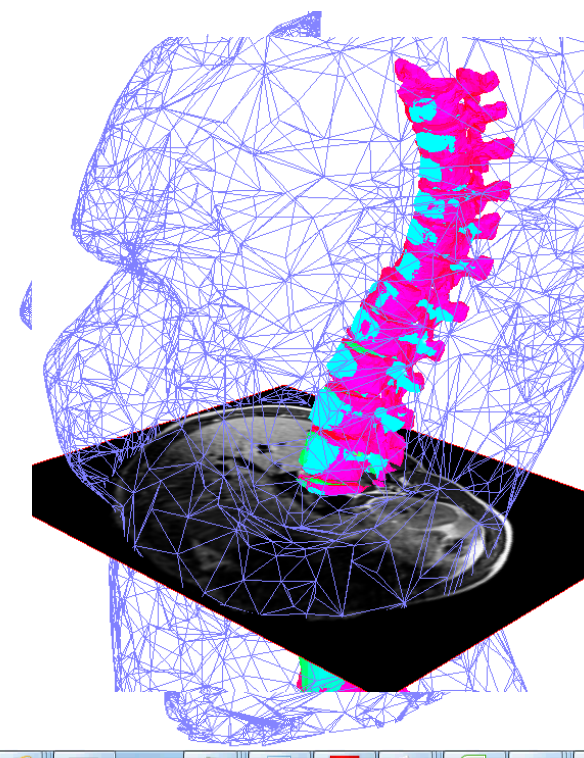
Multimodal image fusion for surgical planning



Rigid registration



**Articulated model
registration**

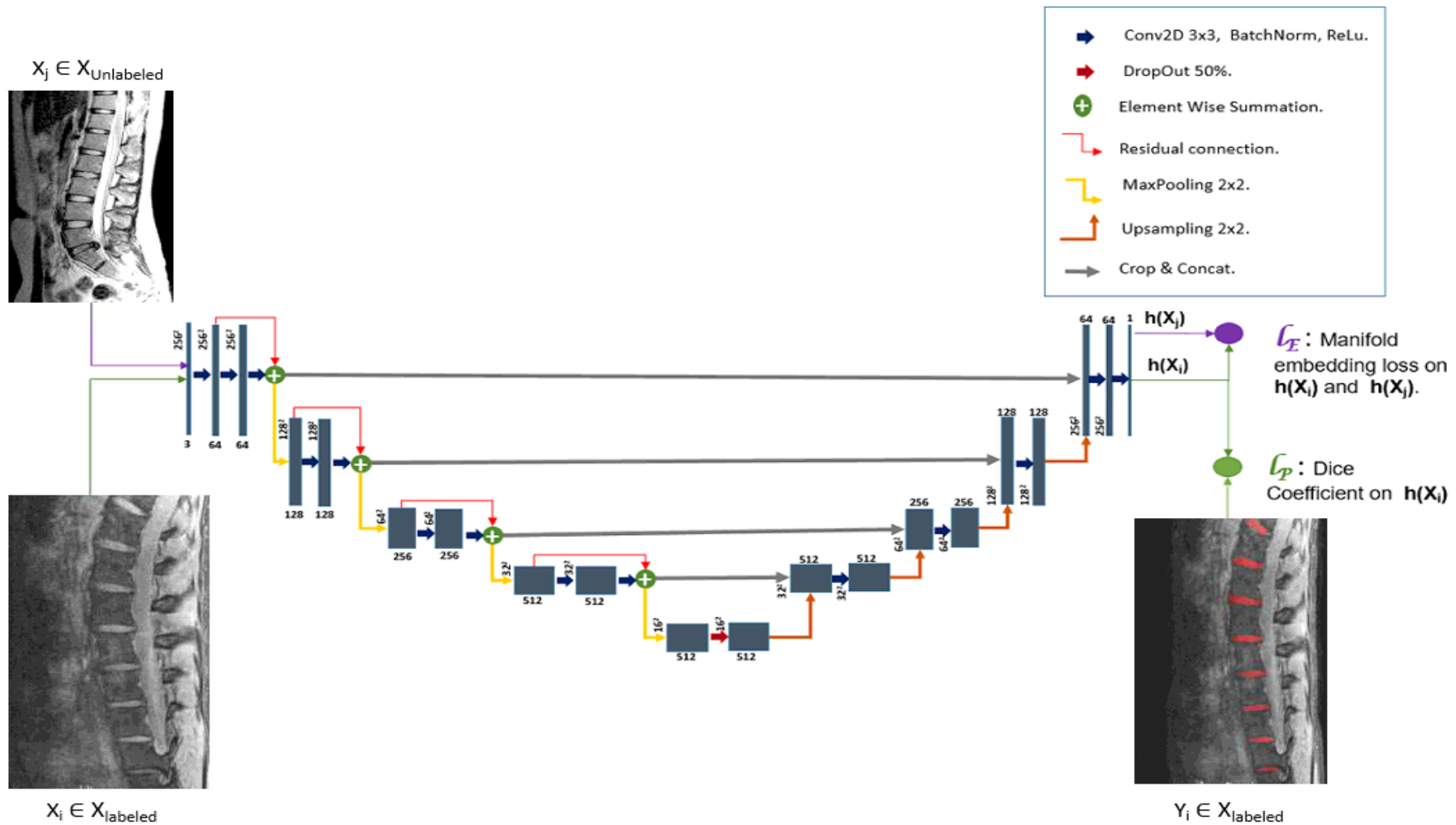


Elastic Registration

Harmouche R, Cheriet F, Labelle H, Dansereau J, Multimodal image registration of the scoliotic torso for surgical planning. BMC Medical Imaging, 2013.



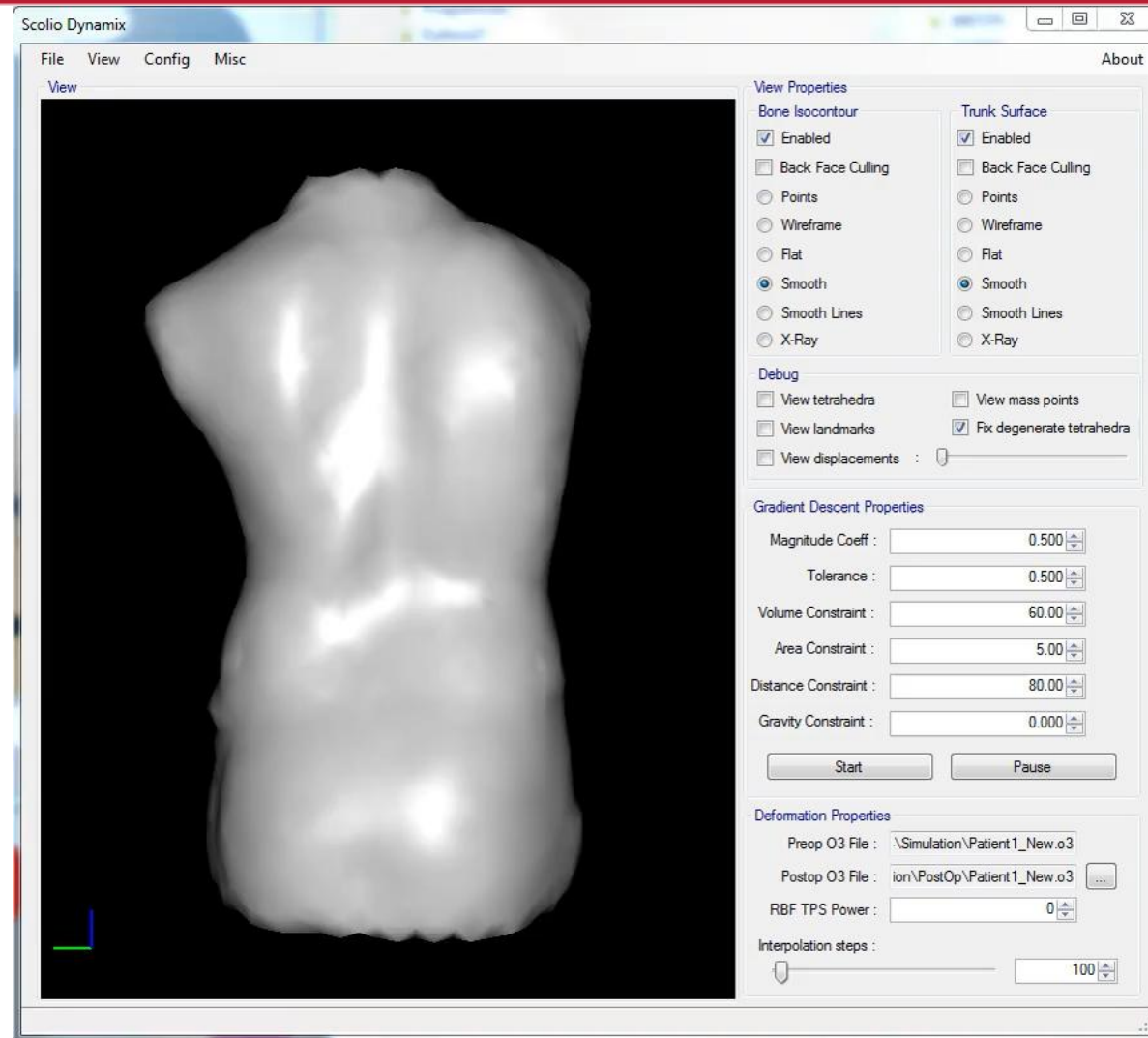
Automatic segmentation of MRI images



Semi-supervised Learning



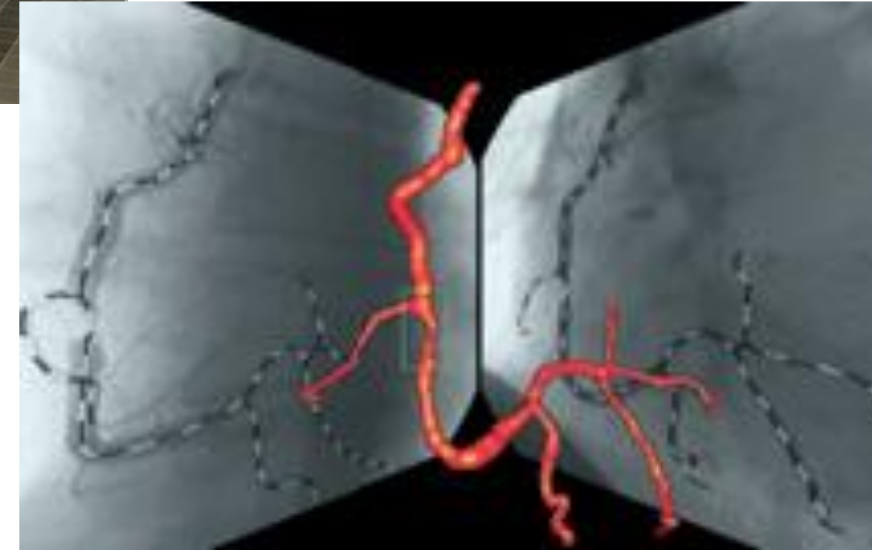
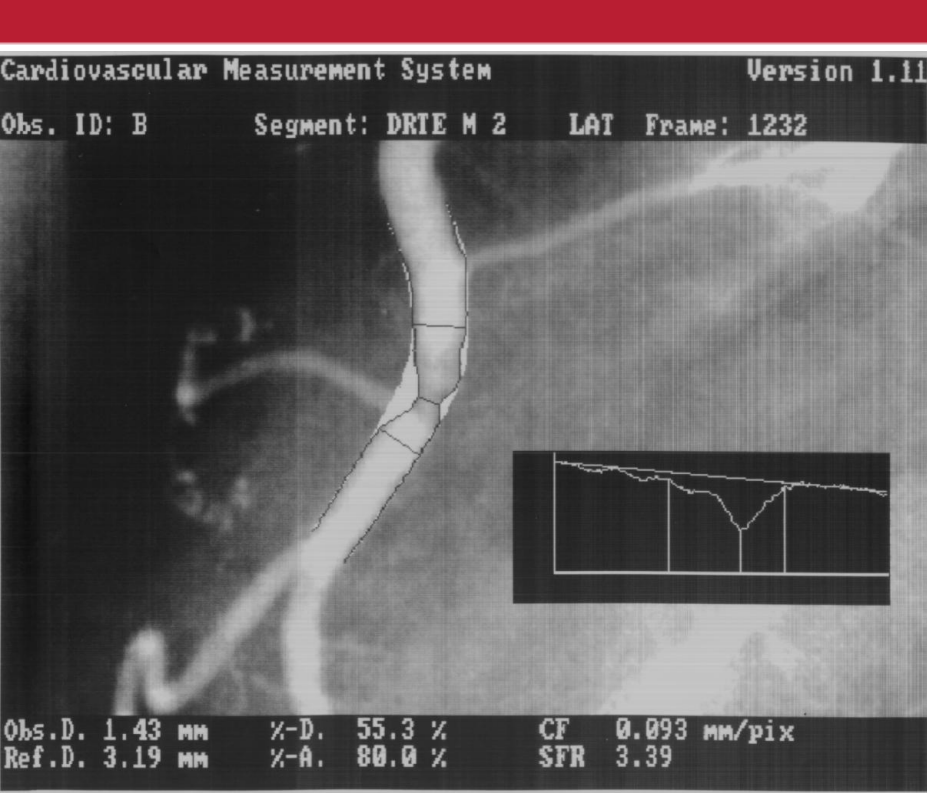
Simulation of postoperative trunk correction



Assi K.C, Grenier S., Parent S., Labelle H., Cheriet F., A physically based trunk soft tissue modeling for scoliosis surgery planning, *Computerized Medical Imaging and Graphics*, 2015.



AI in Cardio-vascular image analysis



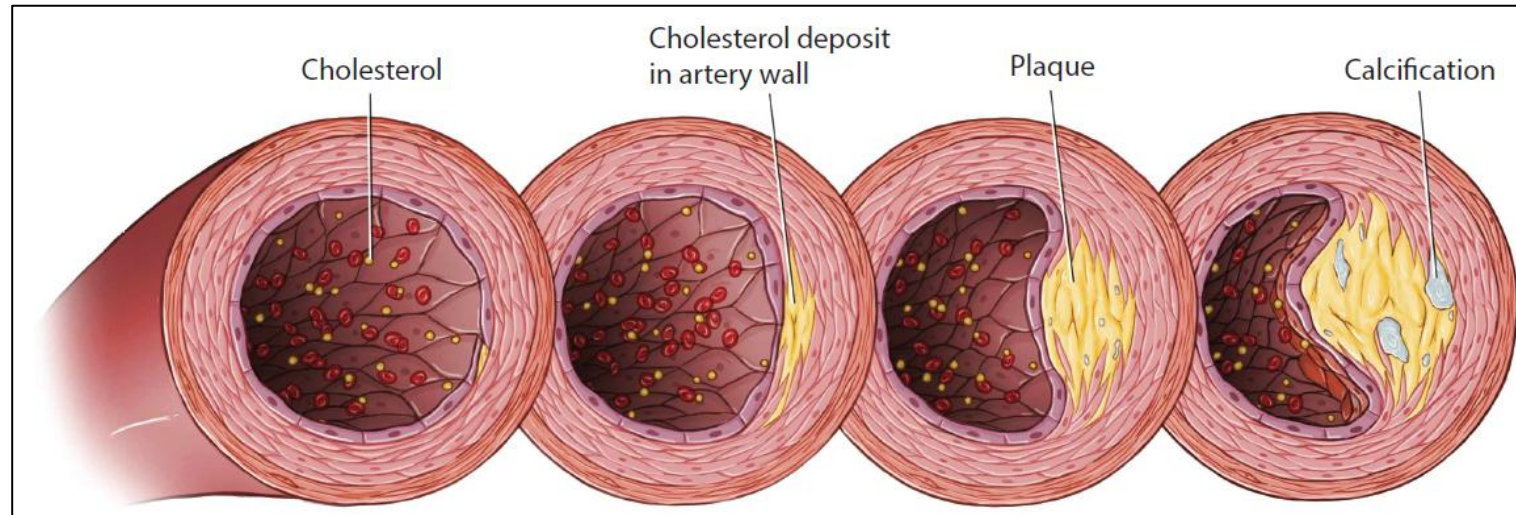
Reiber system used at I.C.M

Fallavollita P, Cheriet F, Optimal 3D Reconstruction of Coronary Arteries for 3D Clinical Assessment, Computerized Medical Imaging and Graphics, 2008.



Coronary artery disease - Atherosclerosis

Atherosclerosis progression to calcified plaque deposits.



Outcomes:

- ↓ lumen size => **angina**
- ↑ wall stiffness
- ↑ thrombosis formation => **infarction**

Coronary artery disease – Kawasaki disease (KD)

- Leading cause of acquired pediatric CAD
- Self-limited acute vasculitis -> CA
- Loss of supportive structures
 - ↓ tunica media
 - ↓ endothelial cells
 - ↓ smooth muscle cells
 - ↓ elastic fibers
 - => Aneurysms in 10%-25% cases.

Precarious fluid mechanics:

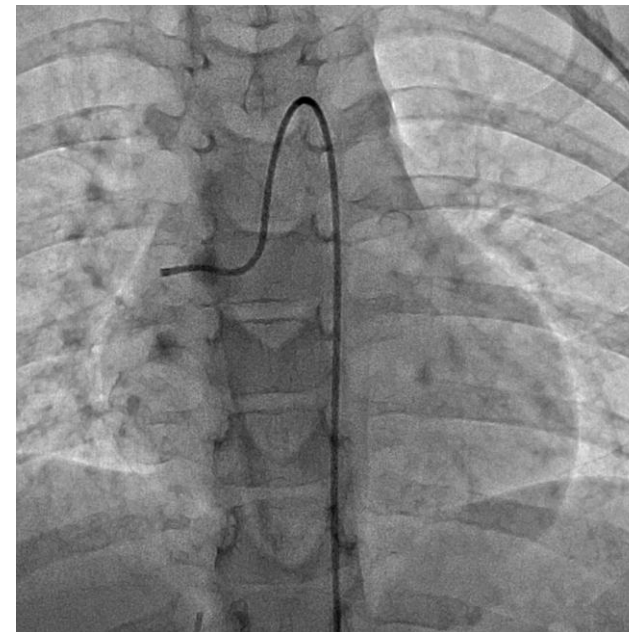
- Platelet activation => thrombosis
- Exercise-induced infarction

Treatment:

- Pharmacological
- No consensus for smaller AN
- Truly healed?



Angio findings – LCA aneurysms & RCA ectasia



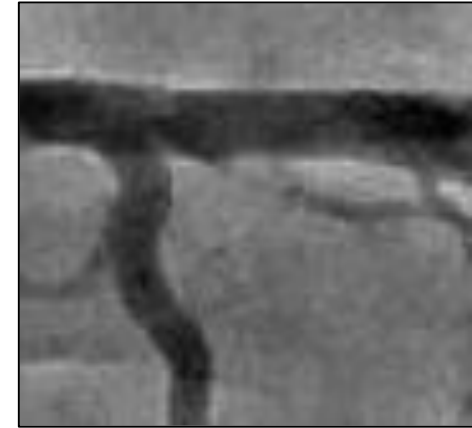
Serial RCA aneurysms



Coronary artery disease – Vascular distensibility

Regressed aneurysms

Endothelium injury
Vascular wall thickening
Calcification



Normal
angiogram (!)

- Intravascular ultrasound (IVUS) + acetylcholine:
impaired vasomotion
- Persistent effect (10+ years)

Atherosclerosis

- Atheromatous changes -> **increased stiffness**
- Non-significant stenosis -> **impaired vasomotion**

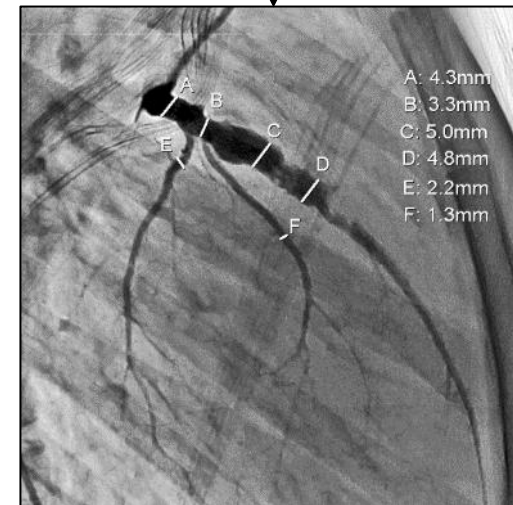


Diagnosis Imaging: X-ray angiography

- 200+ frames per series
- Motion-rich (heart + thorax)

Limitations

- Anatomy appearance is angular dependant
- Collateral vessels are hard to view
 - Ambiguous heart health in certain cases
- Cumulative radiological exposure
 - Genotoxicity
 - Worrisome for pediatric patient
- **Manual, static measures of the CA**
 - Single frame
 - Labor intensive
 - Subjective
 - Large portion of the data ignored



Normal angiographic appearance $\neq$ healthy vascular function

Diagnosis Imaging: Cardiovascular Magnetic Resonance (CMR) Perfusion

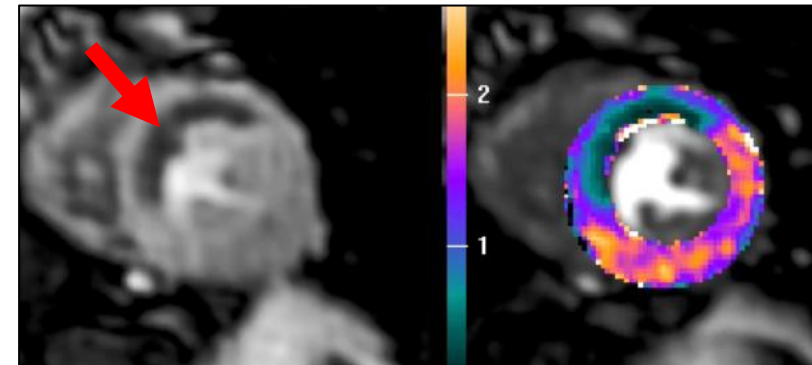
- Non-invasive, non-ionizing
- Detects perfusion defects

Limitations

- Nonrigid and complex motion is prevalent
 - Respiration
 - Voluntary patient displacement
 - Involuntary thoracic organ movement
 - Heart beating
- Strict myocardium alignment is required
 - **Manual** LV, RV, and myo drawing
 - **Very** labor intensive



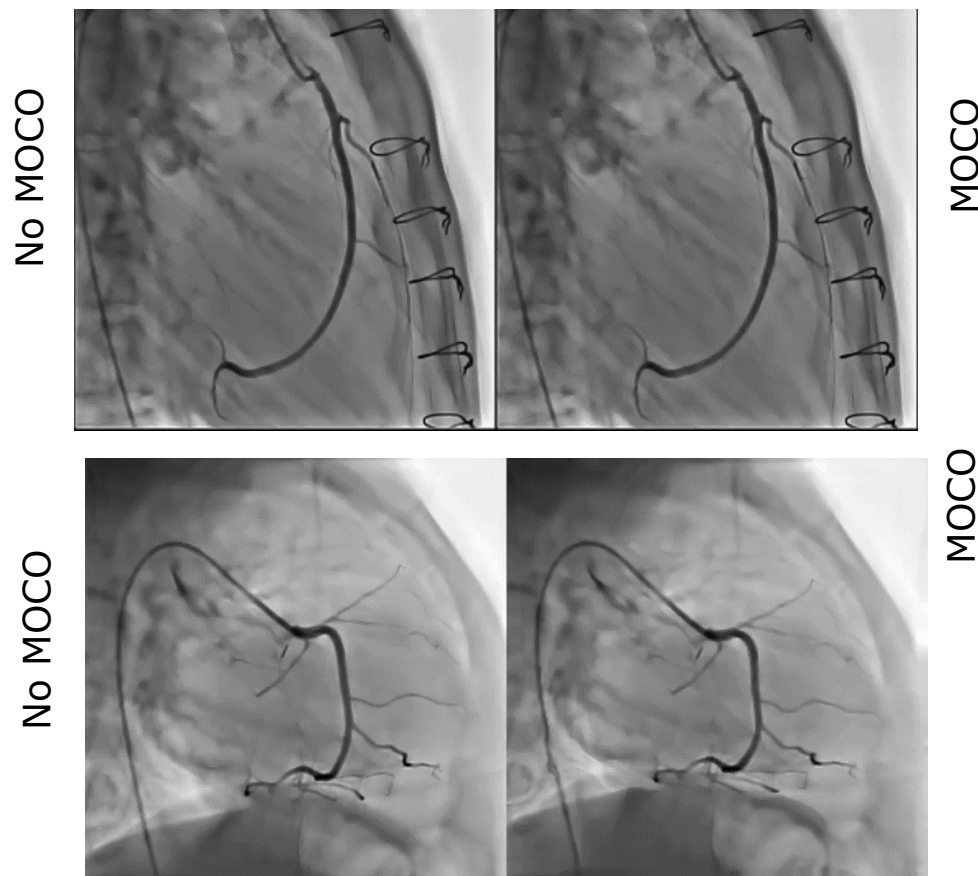
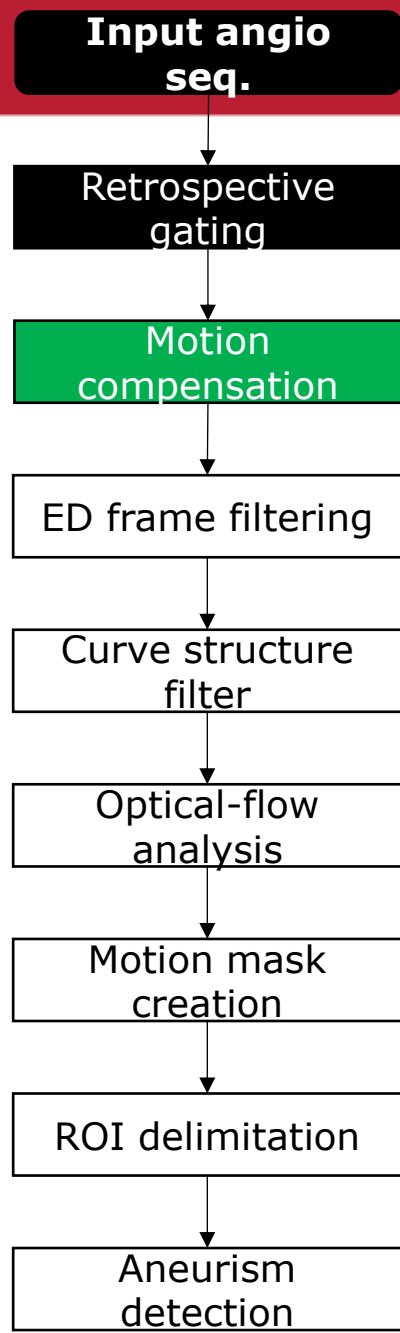
Free breathing perf. series



Infarct patient's quant. perf. pixel mapping



Motion Correction



Benovoy M., Jacobs M., Cheriet F., Dahdah N., Arai A.E., Hsu L.Y., "A Robust Nonrigid Motion Correction Framework for First-Pass Cardiac Magnetic Resonance Perfusion Imaging", *Journal of Magnetic Resonance Imaging*, February 2017.





**Input angio
seq.**

Retrospective
gating

Motion
compensation

ED frame filtering

Curve structure
filter

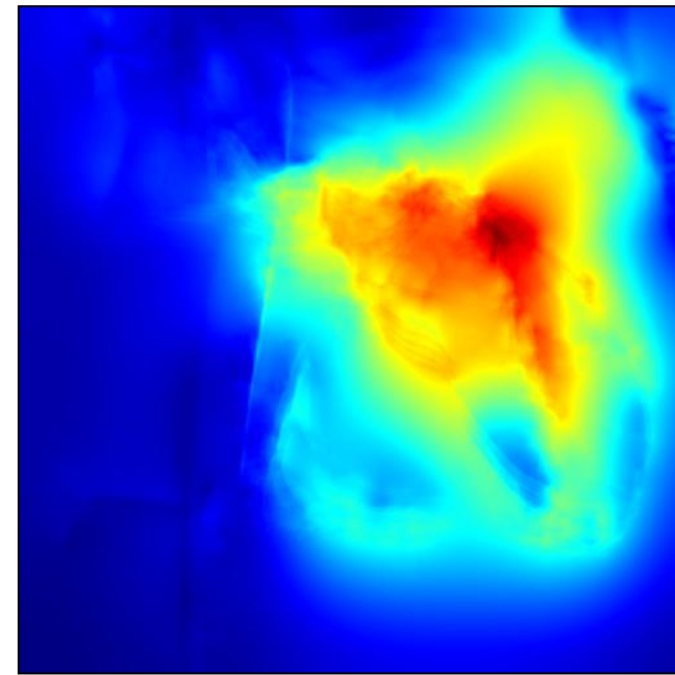
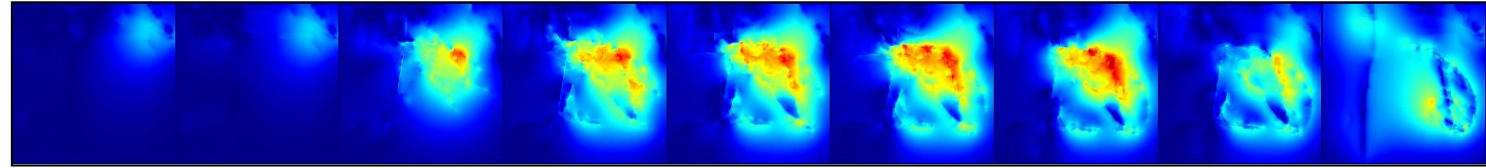
Optical-flow
analysis

Motion mask
creation

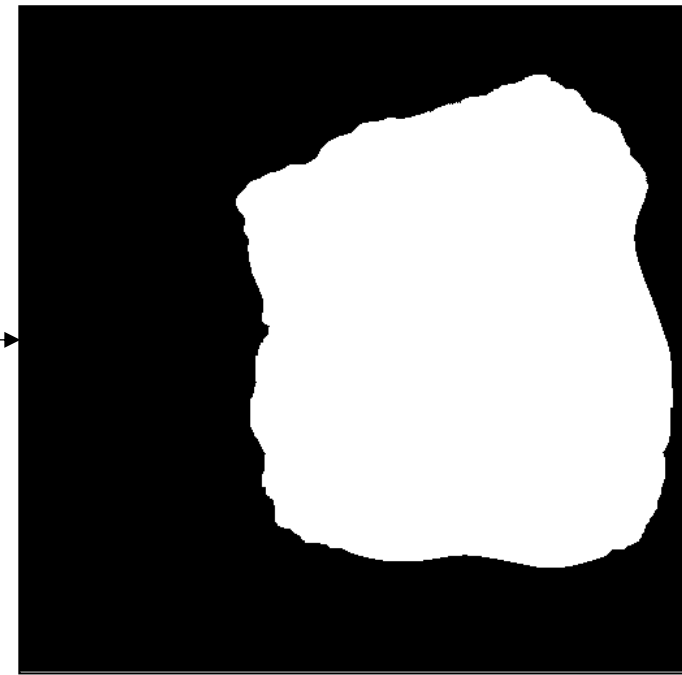
ROI delimitation

Aneurism
detection

End-diastole flow map series



Principal flow map region (kPCA)



Binary motion mask



**Input angio
seq.**

Retrospective
gating

Motion
compensation

ED frame filtering

Curve structure
filter

Optical-flow
analysis

Motion mask
creation

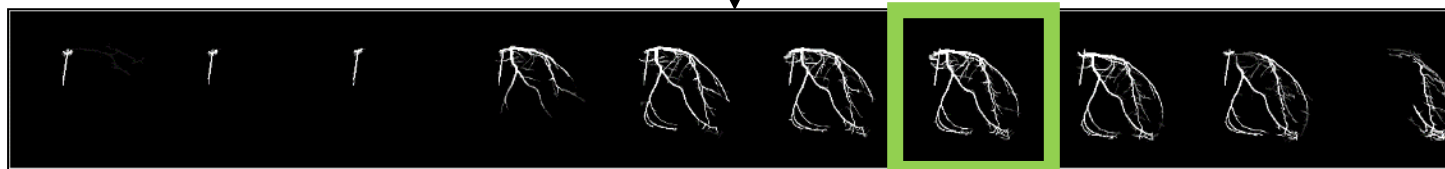
ROI delimitation

Aneurism
detection

End-diastole curve



Binary motion mask replication

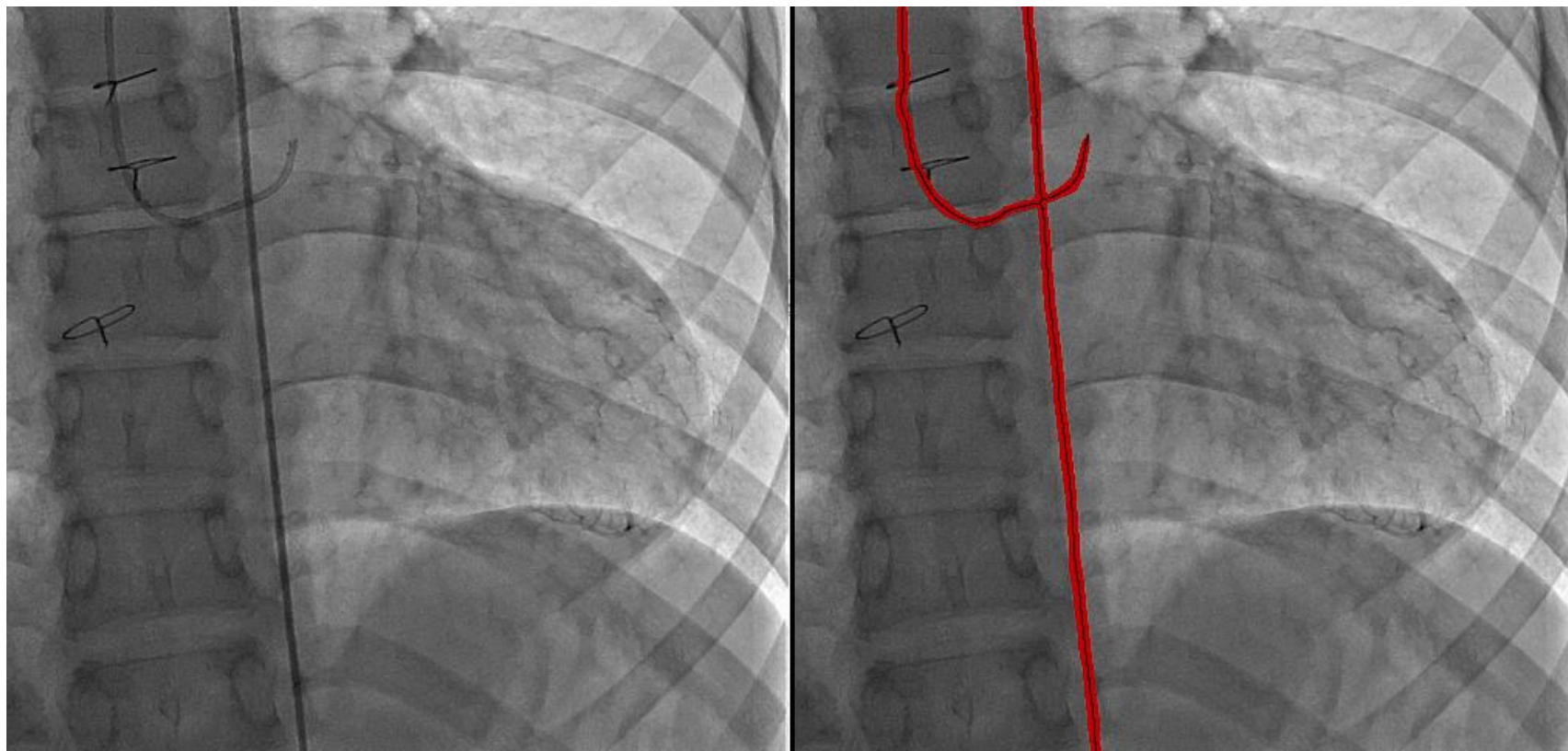
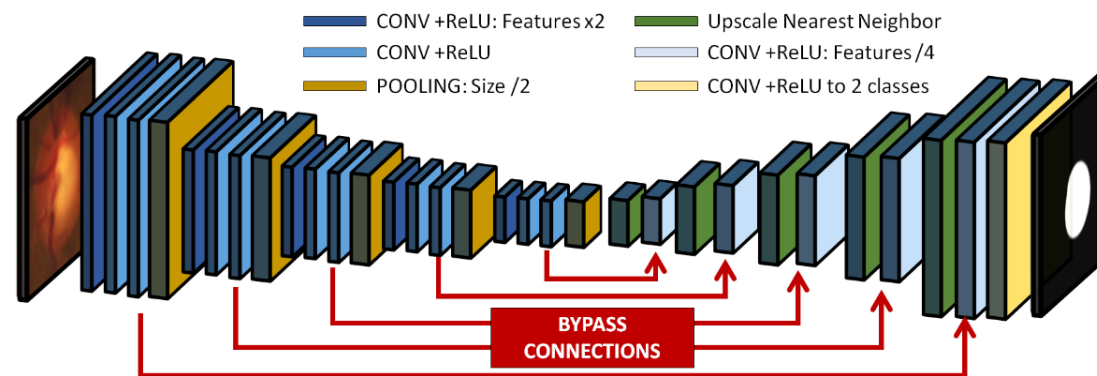


Masked vessel sequence

Mean execution time = 81.03 ± 23.08 s/series (0.38 ± 0.11 s/frame)

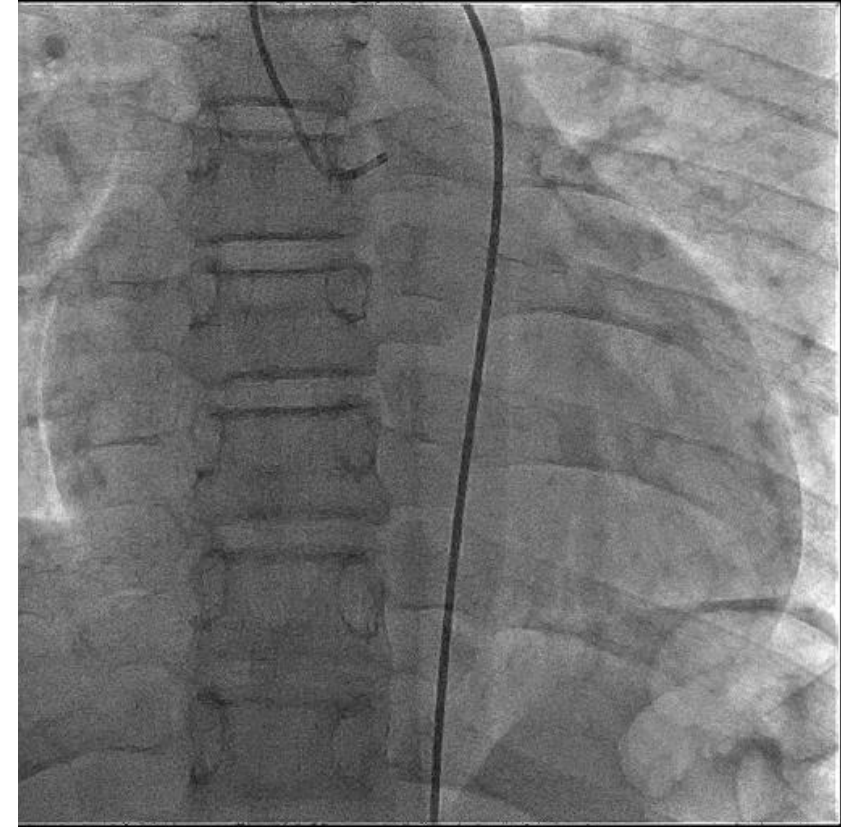
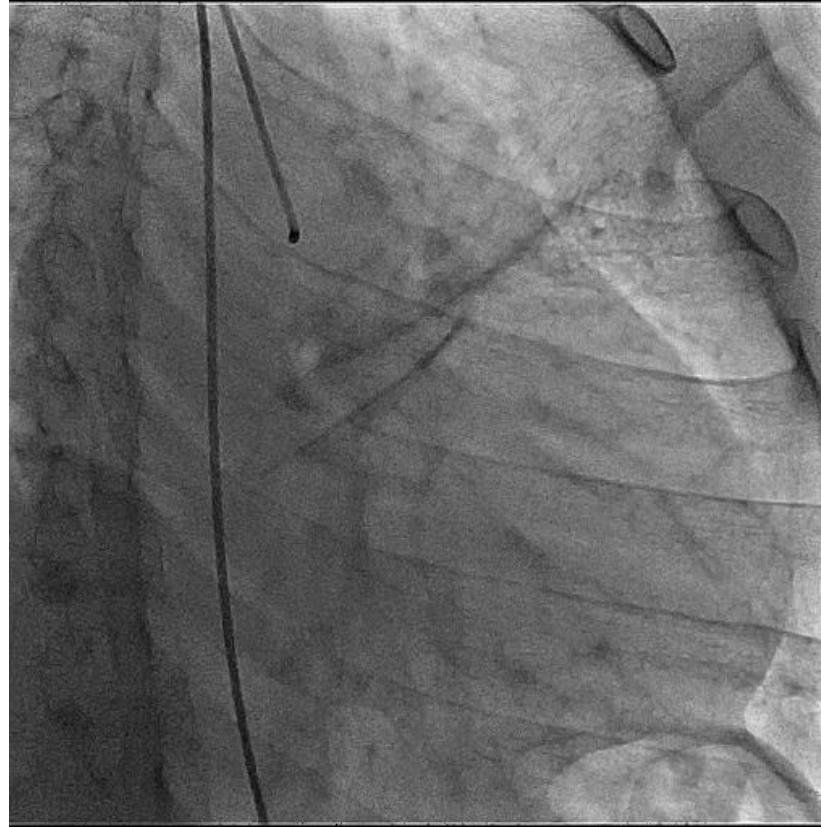


Vessel segmentation using CNN



Temporal Matching

Synchronize
multiple views
of the same heart
based on the
cardiac cycle



Overview of the method

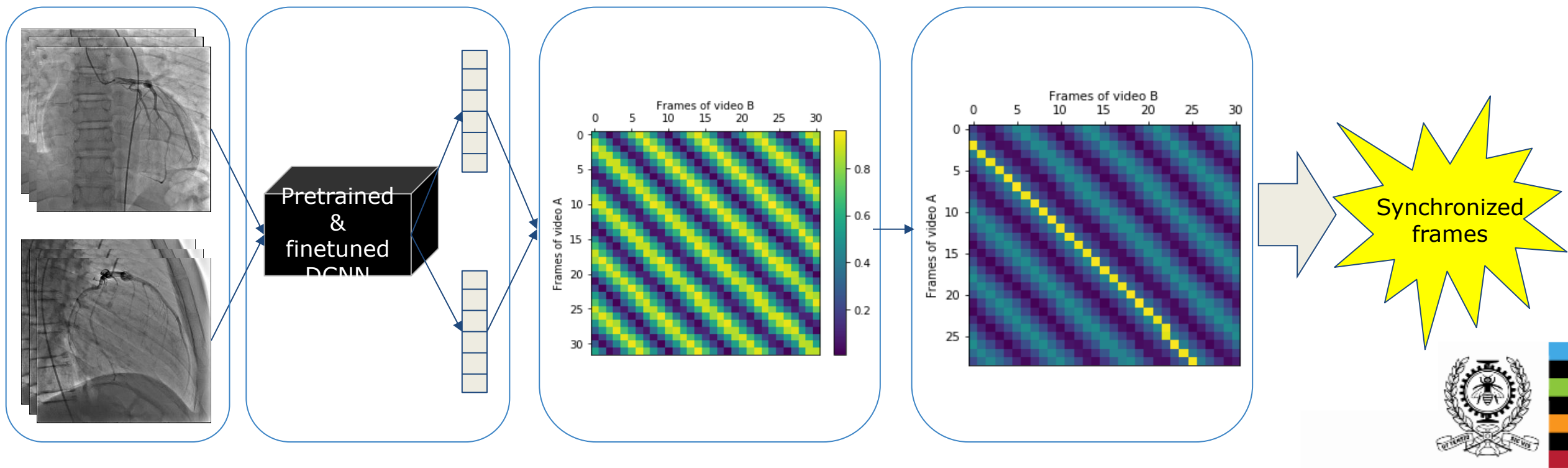
Royer-Rivard R., Girard F., Dahdah N., Cheriet F., End-to-End Deep Learning Model for Cardiac Cycle Synchronization from Multi-view Angiographic Sequences, IEEE EMBC 2020.

1. Sequence extraction

2. Model inference

3. Similarity matrix generation

4. Pathfinding



Results

Here are some examples of synchronized videos pairs of the same coronary artery from different angles.



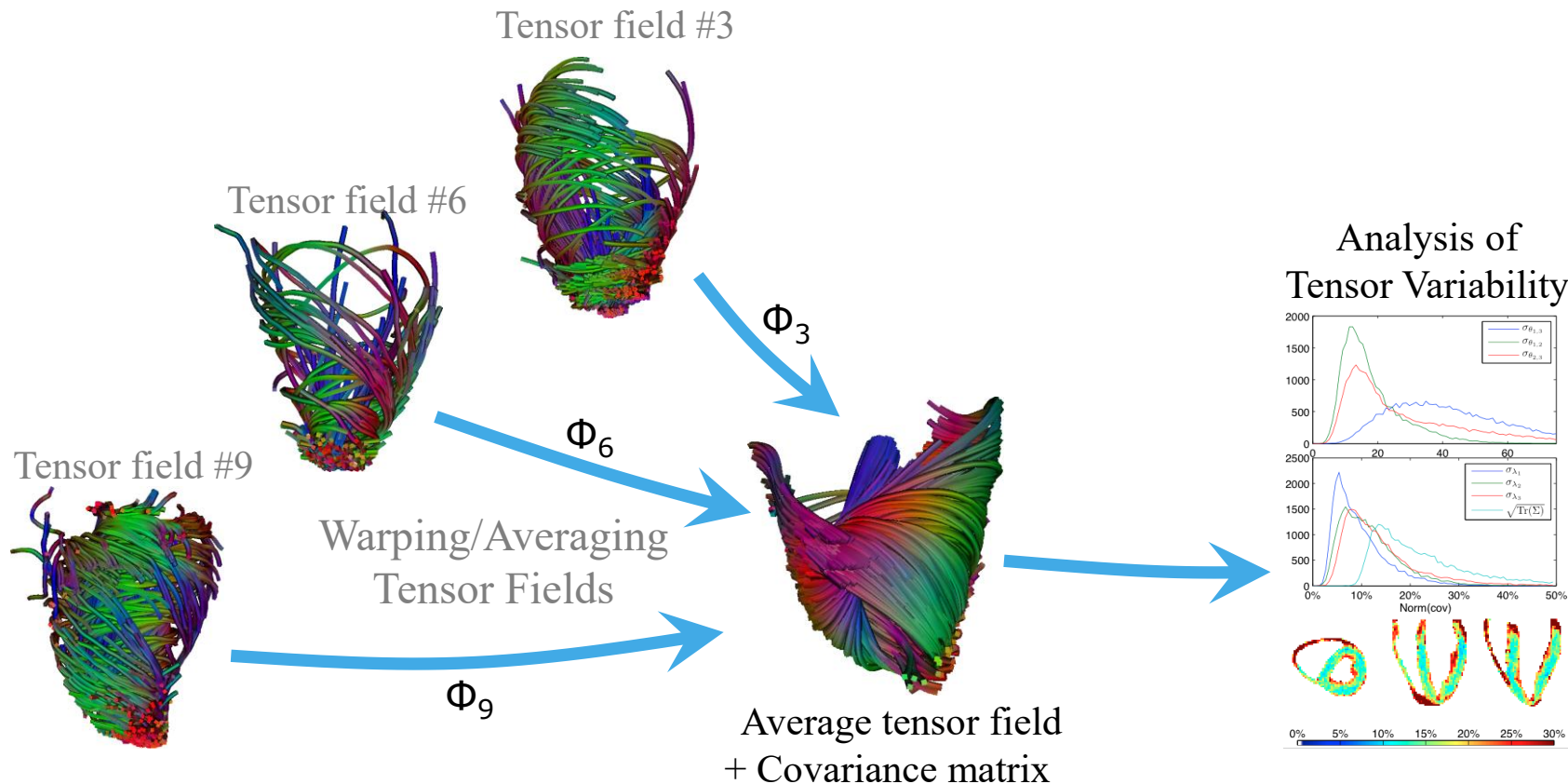
Statistical atlases: Cardiac Fiber atlas

Building the Fiber Atlas

Registrations gave all transformations $\Phi_{1..n}$
Construction of DTI Atlas
by warping all tensor fields with $\Phi_{1..n}$

Two Warping Strategies

1. **Finite Strain Strategy (FS)**
Preserve geometric features (*i.e., inter subject*)
2. Preservation of the Principal Direction (PPD)
Preserve mechanical transformations (*i.e., intra subject*)



Prediction of microvascular pathologies

- Eye as a window to the heart
 - *Nowhere else in the living body are we able to study the blood vessels of the size of the central artery and its branches to the finest arterioles than in the fundus. The man with the ophthalmoscope is impressed by its importance in the recognition of the early signs of general disease and of its general diagnostic and its prognostic value. But its greatest value lies in the possibility it affords to help solve the problems that have so long baffled medical science, in the search for ultimate causes of conditions that are so common, so varied in their manifestations and so serious in their course.*
- Pr. Harry Friedenwald, Oxford 1930



1966

1974

1992

1965

1970

1975

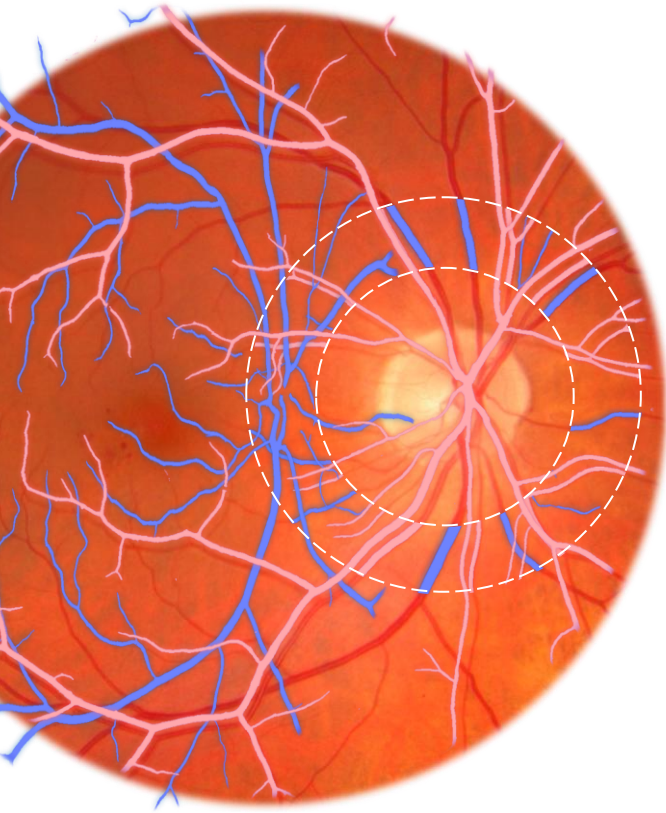
1980

1985

1990

Standardisation of retinal biomarkers

Extraction of Vessel tortuosity, vessel fractal dimension, vessel diameter, AVR (artery/vein ratio)



Kagan, A. *et al.* "A note on signs in the fundus oculi and arterial hypertension: conventional assessment and significance." *Bulletin of the World Health Organization* 34.6 (1966): 955.

J. Parr et G. Spears, "General caliber of the retinal arteries expressed as the equivalent width of the central retinal artery" *American journal of ophthalmology*, vol. 77, no. 4, p. 472–477, 197.

L. D. Hubbard *et al.* "Methods for evaluation of retinal microvascular abnormalities associated with hypertension / sclerosis in the atherosclerosis risk in communities' study" *Ophthalmology*, vol. 106, no. 12, p. 2269–2280, déc. 1999.



AI in retinal image analysis

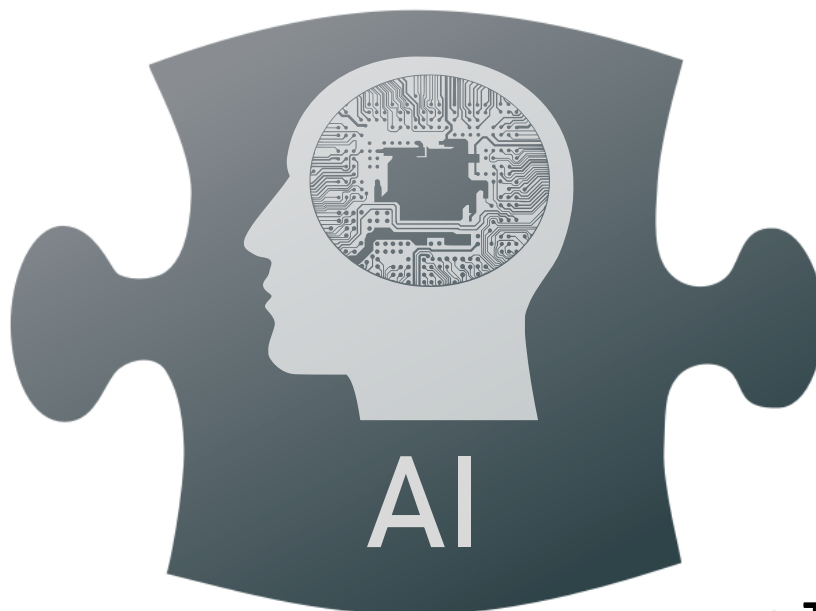
- Ocular diseases:
 - Diabetic Retinopathy (DR);
 - Age-related Macular Degeneration (AMD);
 - Glaucoma
- **Non-invasif** imaging of the eye.



Enabling DR early screening through AI



- **Up to 40% of Canadians with diabetes are not screened for DR.**
- The number of diabetic cases is continually increasing.

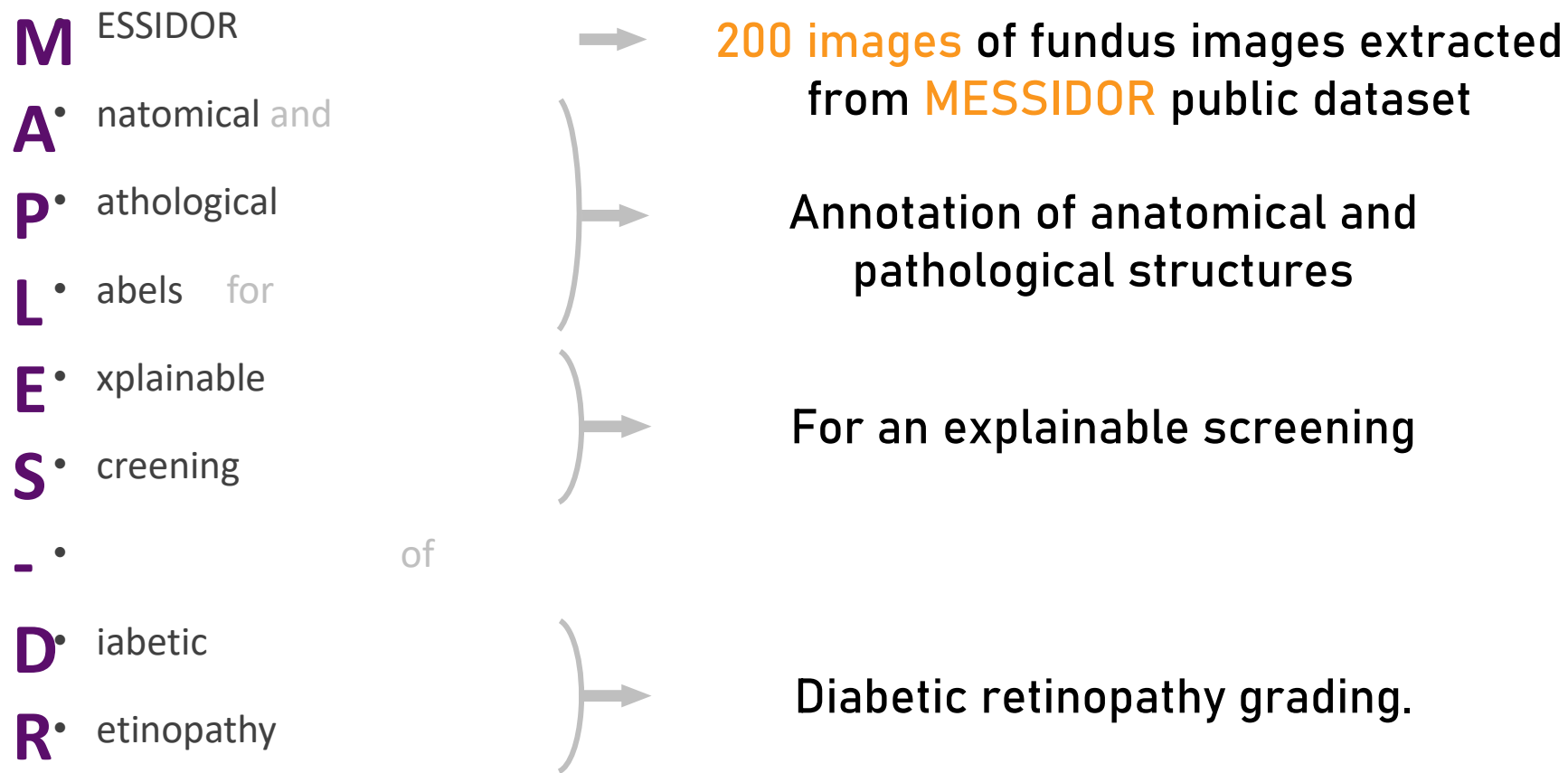


- **Telemedicine Networks** are emerging as a promising solution...
- ... but their capacity is limited by the number of trained readers.

⇒ **Artificial Intelligence (AI)** can facilitate early screening of DR cases and facilitate access to retinal specialists to patients who need care.



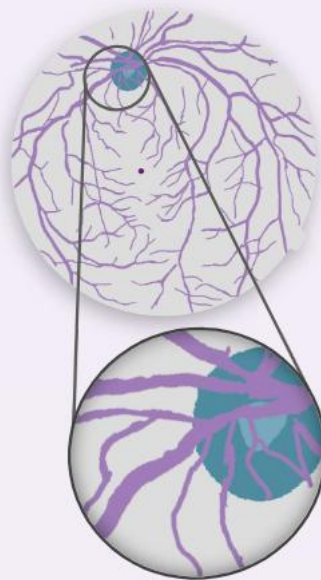
MAPLES-DR dataset



MAPLES-DR annotation

200 x

4 ANATOMICAL STRUCTURES



- Optic Disc
- Optic Cup
- Macula
- Vessels

6 PATHOLOGICAL STRUCTURES



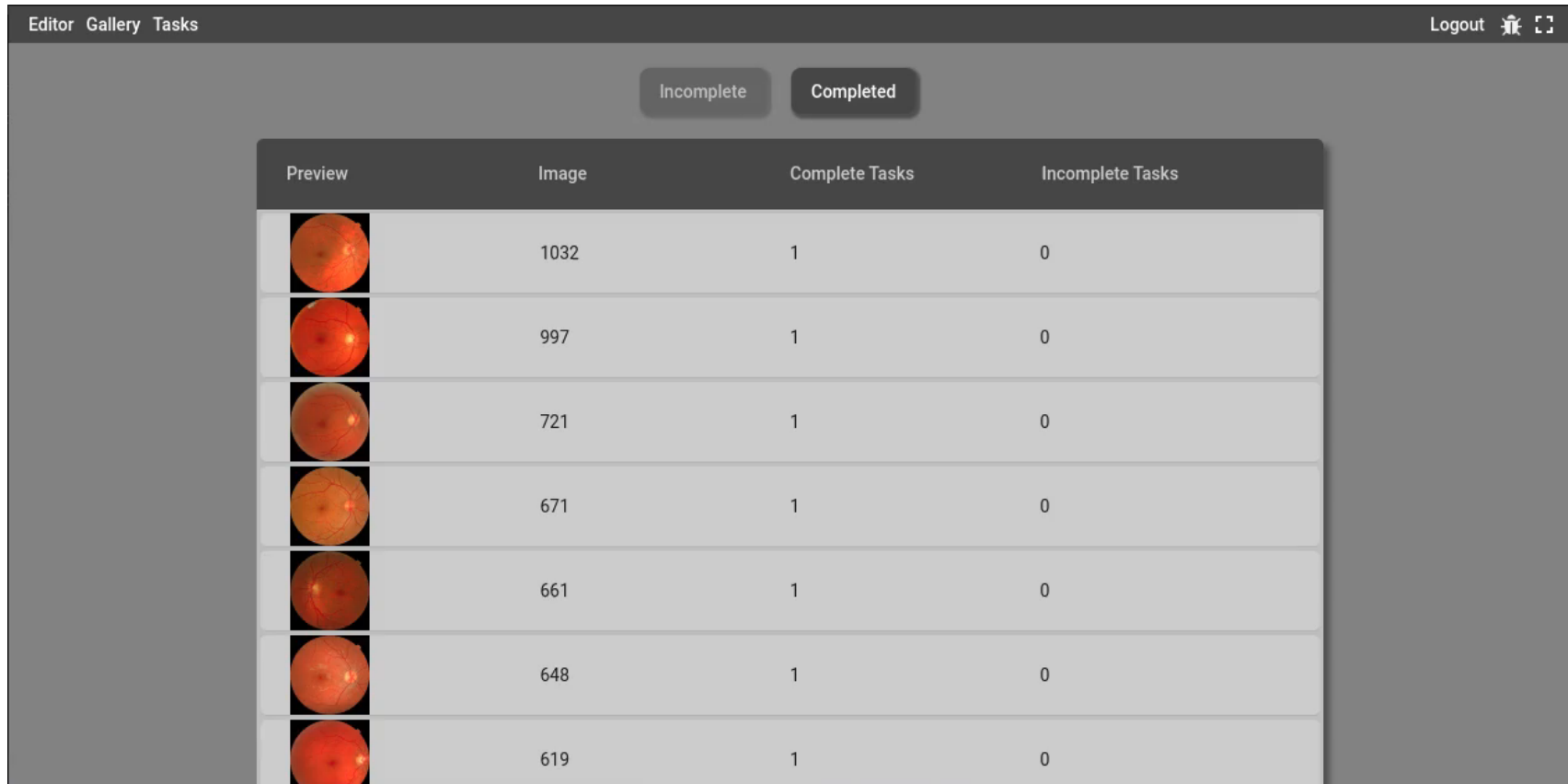
- Exudates
- Cotton Wool Spots
- Drusens



- Micro Aneurysms
- Hemorrhages
- Neo-vessels

⇒ Annotated by 7 **rétinologues** from Québec and d'Ontario.

Annotation Plateforme



The screenshot shows the 'Tasks' tab of the annotation platform. At the top, there are tabs for 'Editor', 'Gallery', and 'Tasks'. On the right, there is a 'Logout' button and a user icon. Below these, there are two buttons: 'Incomplete' and 'Completed'. The main content is a table with four columns: 'Preview', 'Image', 'Complete Tasks', and 'Incomplete Tasks'. The table lists seven tasks, each with a retinal fundus image preview, an ID number, and task completion counts.

Preview	Image	Complete Tasks	Incomplete Tasks
	1032	1	0
	997	1	0
	721	1	0
	671	1	0
	661	1	0
	648	1	0
	619	1	0

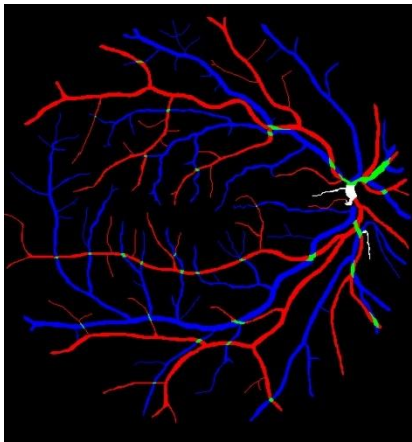
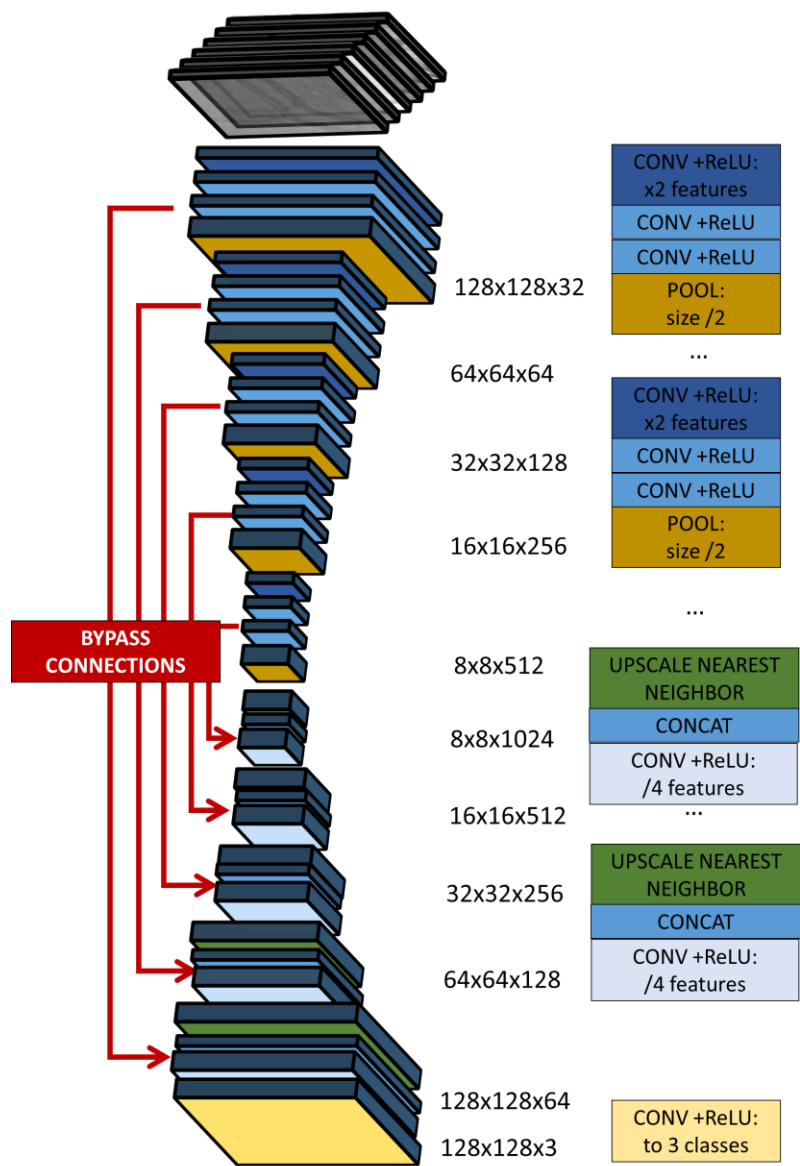
 on ligne

 Multi-users

 Support for graphical
tablette

Lepetit-Aimon, G., Playout, C., Boucher, M. C., Duval, R., Brent, M. H., & Cheriet, F.
" **MAPLES-DR: MESSIDOR Anatomical and Pathological Labels for Explainable Screening of Diabetic Retinopathy.** " *Scientific Data* 11.1 (2024) : 914.

segmentation/classification of veins and arteries

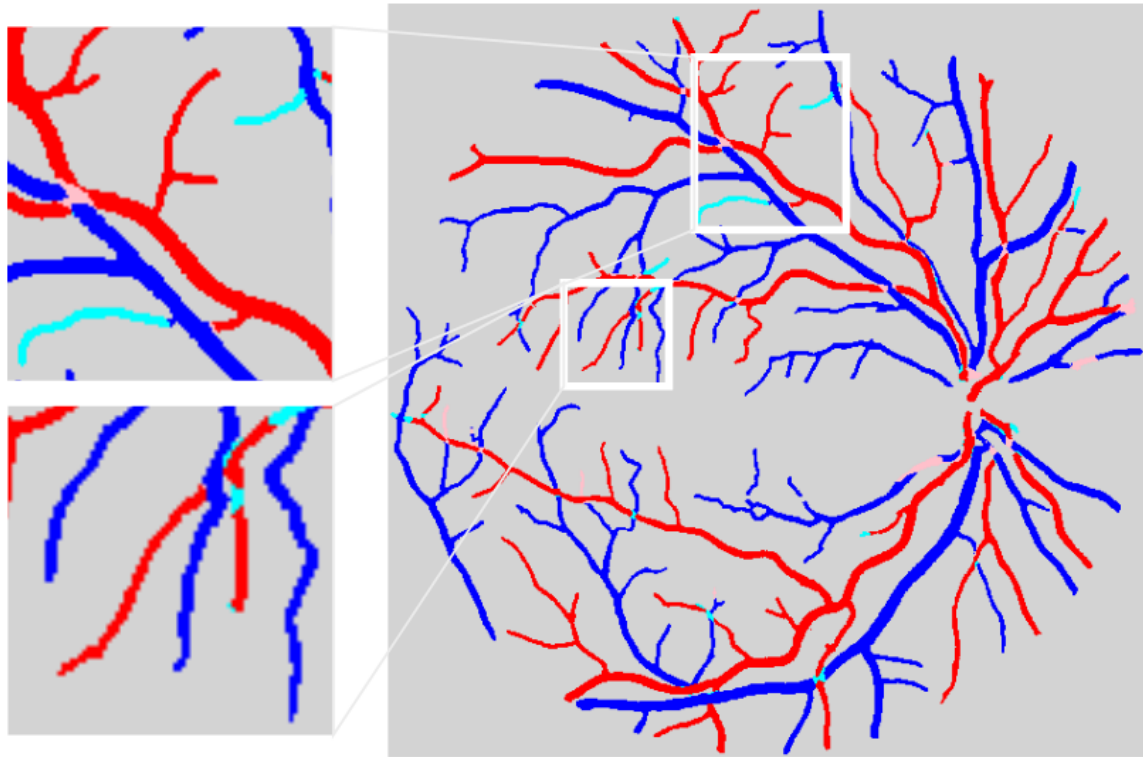


Method	Sensitivity	Specificity	Time
CNN 6D BP + LSP	93.7% ± 4%	92.9% ± 5%	0.5s
Estrada et al.[32]*	91.7% ± 7%	91.7% ± 7%	131.2s
Dashbozorg et al.[29]	90.0%	84.0%	-
CNN 6D BP	88.1% ± 5%	85.7% ± 7%	0.4s
Niemeijer et al.[5]	80.0%	80.0%	-

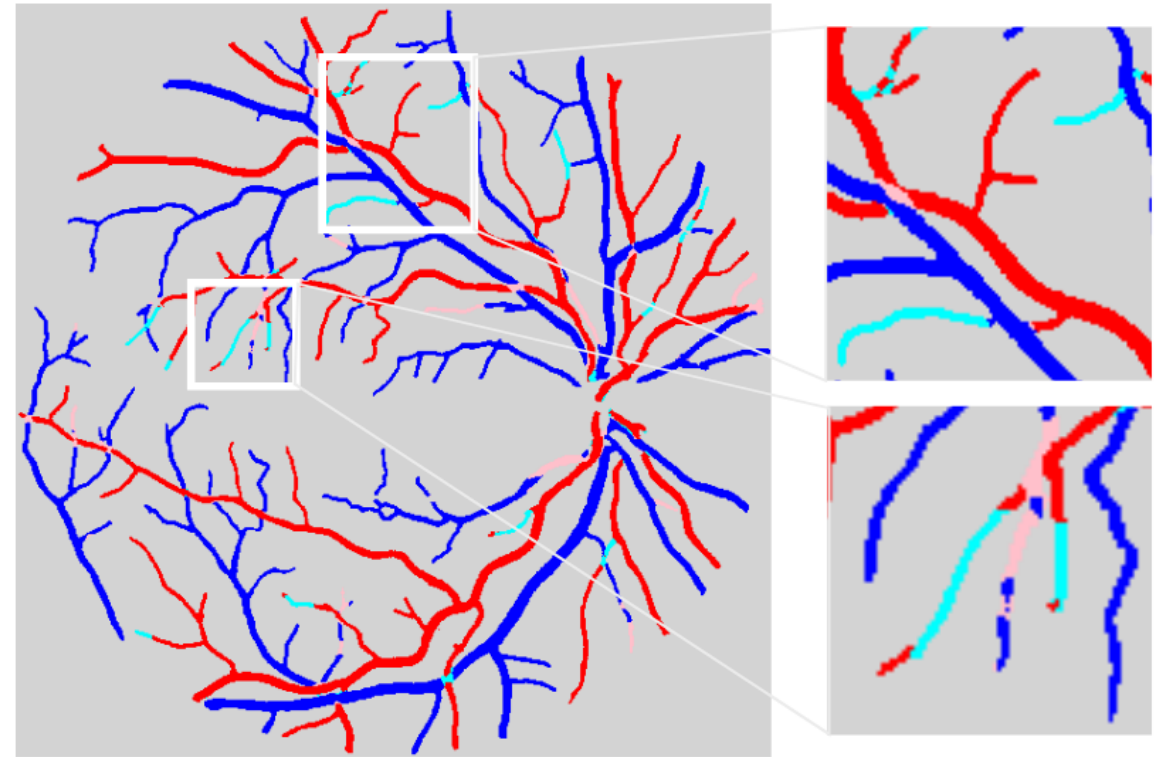
- Girard F., Kavalec C., Cheriet F., Joint Segmentation and Classification of Retinal Arteries/Veins from Fundus Images, Artificial Intelligence in medicine, 94:96-109, 2019.



Artery/Vein classification



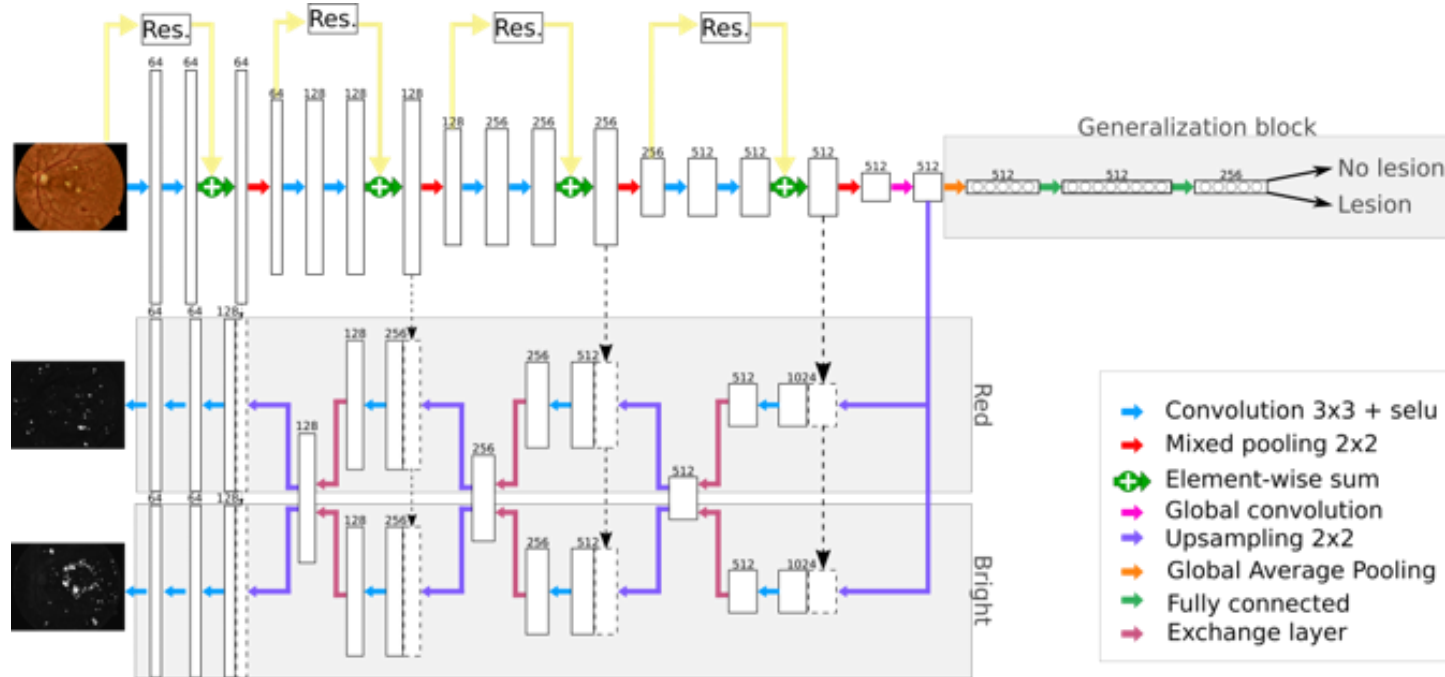
STEERED CNN



STANDARD CNN

Multitasking and weak supervision for retinal lesions segmentation

Lesions in the retina are indicators of the presence of a disease as well as its severity and progression.



- Pixel-level clinical annotations are rare. We propose an approach that combines supervised learning on a small annotated database at a pixel-level and weakly-supervised training using images annotated at a global level (presence or absence of disease).
- Our network is composed of three branches (ie. three tasks).
 - Generalization block*: image-level diagnosis
 - Red lesions*: hemorrhages and microaneurysms
 - Bright lesions*: exudates and drusen.



Explainable/Interpretable AI

- Most deep learning decision systems are considered as a black box
 - Convolutional Neural Networks for disease classification
 - Visualization of a heatmap for explainable decision.
- Recently (2020/2021), (Vision) Transformers were adapted for image classification
 - Since, ViTs are getting more and more popular
- We propose a new mechanism to improve the performance and the interpretability of the prediction of a ViT:

Clément Ployart, Renaud Duval, Marie Carole Boucher, Farida Cheriet, ***Focused Attention in Transformers for interpretable classification of retinal images,***
Medical Image Analysis, Volume 82, November 2022



Vision Transformer: how does it work?

Image as sequence

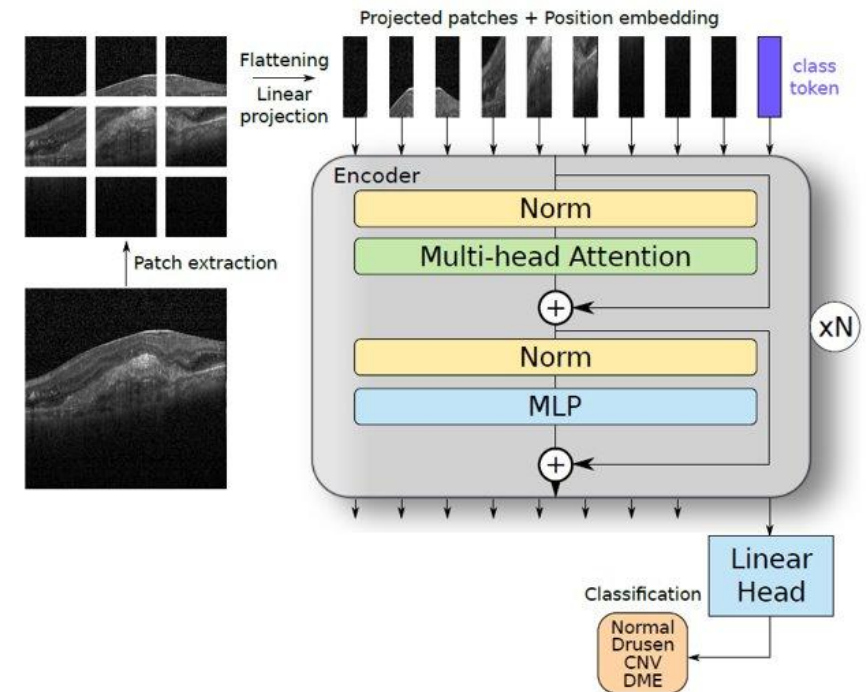
- The image is decomposed in patches
- Features are extracted from patches
- Features form a 1D sequence of **tokens (1 token = 1 patch)**
- A **classification token** is added to the sequence



Building an architecture

As for CNN, by stacking layers:

- A layer of ViT is composed of different steps:
 - Normalization, linear projection heads, global attention and multi-layers perceptron
- **In an actual architecture, these layers are stacked**, recombining the sequence layer per layer
- In the last layer, the **classification token** is projected to the actual classes we want to predict



Locality vs complexity

Problem: Retinal diseases may depend on tiny localized structure:

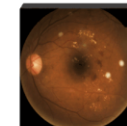
Eg: Microaneurysms in fundus are early indicators of DR but are only a few pixels wide.

1. Patches too large don't provide insights on small lesions

2. Reducing the patches size increases their number and makes the attention computation prohibitive



Large patches are non-informative (global context)



Small patches leads to large attention matrix (and computation intractable with regular hardware)

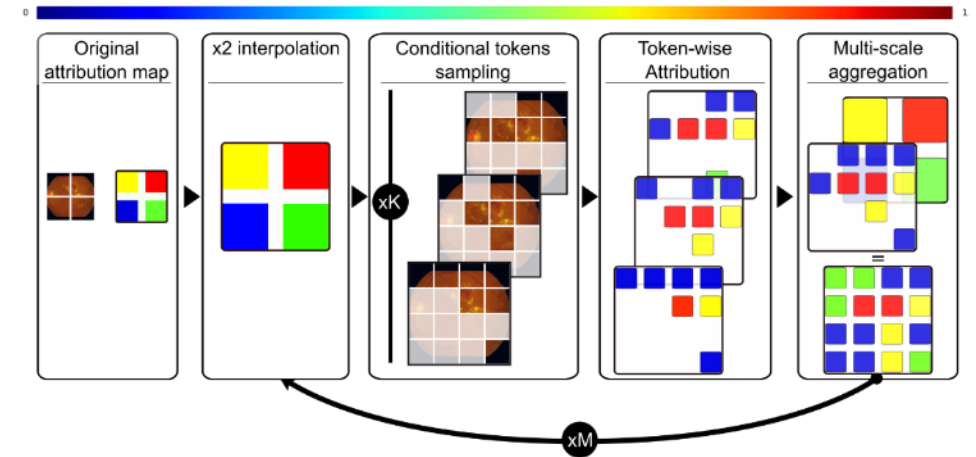


Our approach: Focused Attention

Focused Attention is an iterative approach:

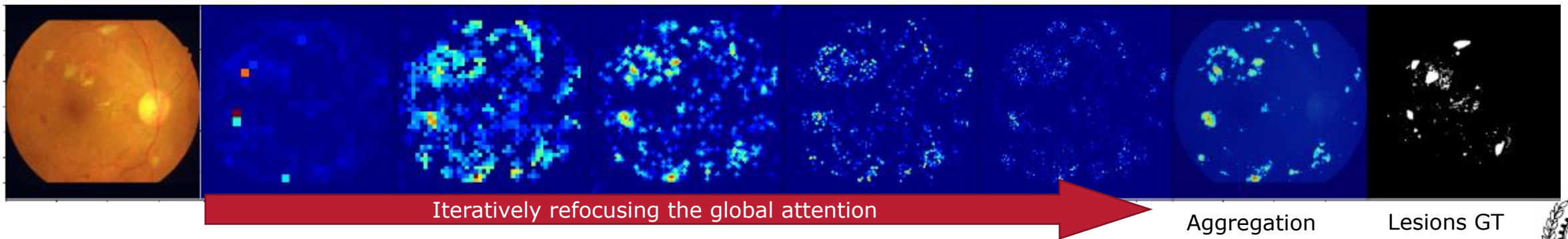
*Instead of resampling patches uniformly, it resamples them in the regions of interest (**ROI**)*

1. The ROI is obtained by looking at the most important patches (highest affinity with the classification token) at the previous iteration
2. In the ROI, patches are resampled at highest resolution **stochastically**
3. At the following iteration, the attention is only computed on the sampled patches



Focusing at different scales

- The iterative approach looks for the most important patches at different resolution.
 - *In practice, we used either Attention Rollout or ViT-Layer Relevance Propagation to generate the attributions maps at each resolution*
- The most informative patches are aggregated at different scales:
 - *Combining mean and max per scale.*
- The final map reliably predicts lesions (with respect to a groundtruth) in a weakly-supervised way (**only the disease is known**)



Classification Performance

- We conduct experiments on:
 - Fundus images for Diabetic Retinopathy grading
 - SD-OCT (macula-centered B-scan) for identification of Diabetic Macular Edema, Drusen and Choroidal Neovascularization
- Vision Transformers obtained performance similar to state-of-the-art approaches (CNNs):

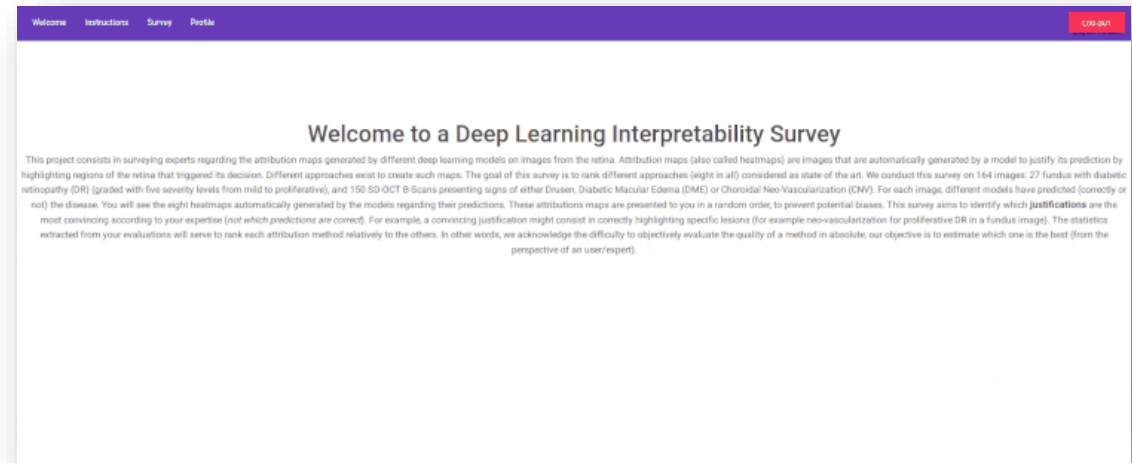
Models	Accuracy	Precision	Sensitivity	Specificity	#params(M)	MACs(G)
Baseline models						
Wide ResNet101	97.5	95.7	95.0	98.3	477	124.85
Optic-Net71	98.1	96.6	96.2	98.7	48	12.45
ResNet152	98.5	97.2	97.0	99.0	223	58.15
Transformer-based models						
ViT _{base} ⁽³²⁾	96.7	94.6	93.4	97.8	87	10.98
ViT _{base} ⁽¹⁶⁾	95.8	93.6	91.6	97.2	86	47.31
DeiT _{base} ⁽¹⁶⁾	97.7	96.0	95.3	98.4	86	47.40
ViT _{large} ⁽¹⁶⁾	99.0	98.0	97.9	99.3	303	162.0
T2T-ViT _{base}	95.6	93.2	91.11	97.0	21	13.89
T2T-ViT _{large}	96.3	93.5	92.6	97.5	63	38.03

Performance evaluated on 1000 B-scans for the identification of images with DME, Drusen or CNV

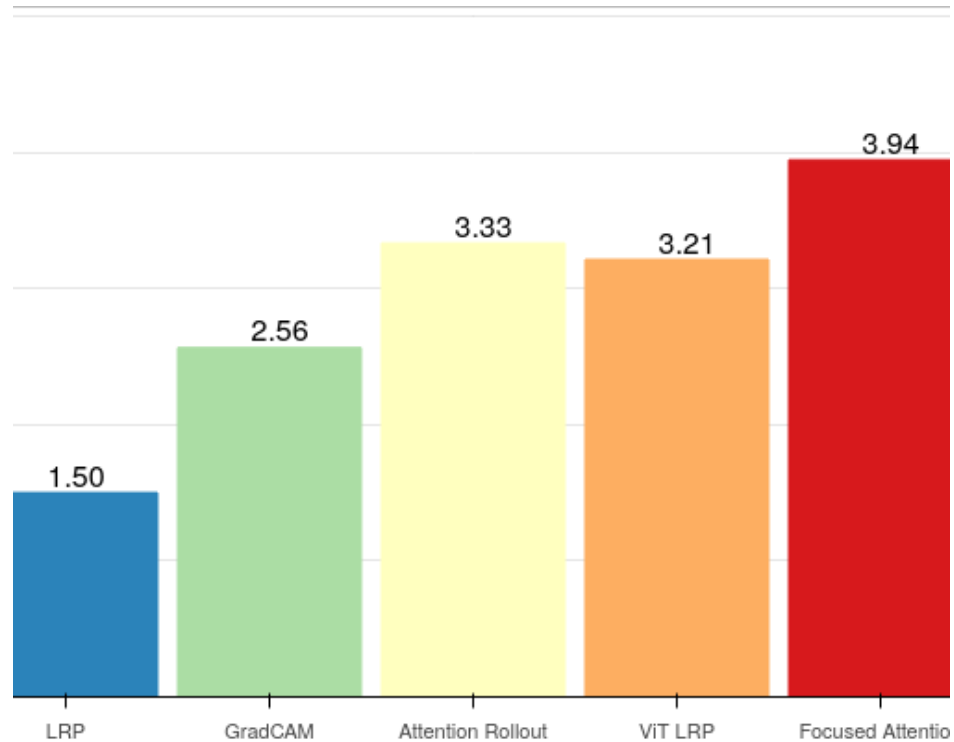


Validating the interpretability

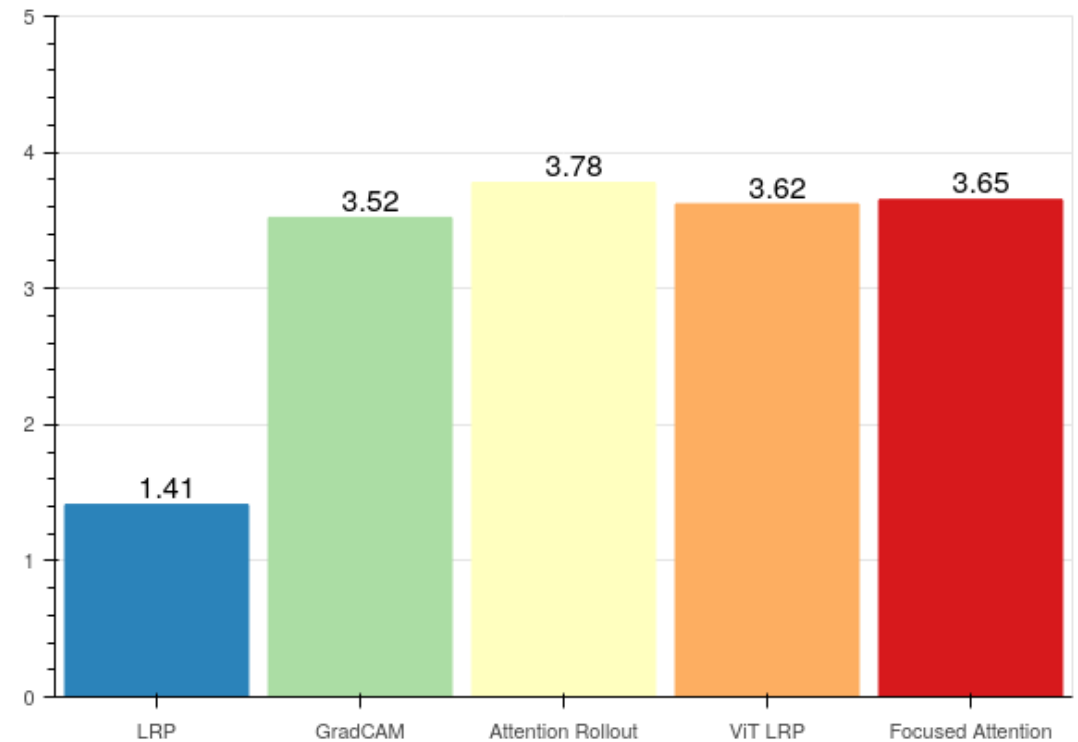
- We conceived an online tool for validating the interpretability:
 - 4 experienced ophthalmologists were given with 200 images of fundus and OCT to review
 - They were asked to grade the attribution heatmaps generated by eight methods (state of the art + focused attention)



Survey results



Fundus



OCT

On fundus, Focused Attention obtained the highest approval from experts (3,94/5). On OCT, it was the second-best preferred method.



Conclusion

- **State of the art algorithms underperform on new datasets:**
 - Unlabeled datasets and unbalanced classes in clinical environments.
 - Public datasets are not always reliable and have license restrictions limiting their usage.
 - Generalization of learned models to new datasets is still an open problem.
 - **We are exploring technical solutions to overcome these challenges:**
 - Investigating self-supervised approaches.
 - Enhancing algorithm reliability by using quality scores and multi-view prediction.
 - Investigating interpretable models.
- ⇒ **Collaboration between AI researchers and medical specialists is essential to identify, understand and overcome these challenges!**





Canada

CIHR IRSC

Canadian Institutes of Health Research
Instituts de recherche en santé du Canada



Diabetes
Action
Canada

Preventing complications.
Transforming lives.

Université 
de Montréal

The University of Montreal
and of the world.



CRSNG
NSERC

Canada

Natural Sciences and Engineering
Research Council of Canada

Conseil de recherches en sciences
naturelles et en génie du Canada